

Affect First!?

Evaluating the Prominence of Affective Meaning through Valence

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Institut | Nicod



| PSL 



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Affect-First Hypothesis

Wundt 1907; Zajonc 1980, 2000; Murphy & Zajonc 1993; LeDoux 1996; Stapel, Koomen & Ruys 2002

"You d*mb B*TCH!"

"Eat me."



"You d*mb B*TCH!"

"Hwhyyyyy?"



Roadmap

A. Prelude

B. Affect-First Hypothesis

C. Conceptual Domains

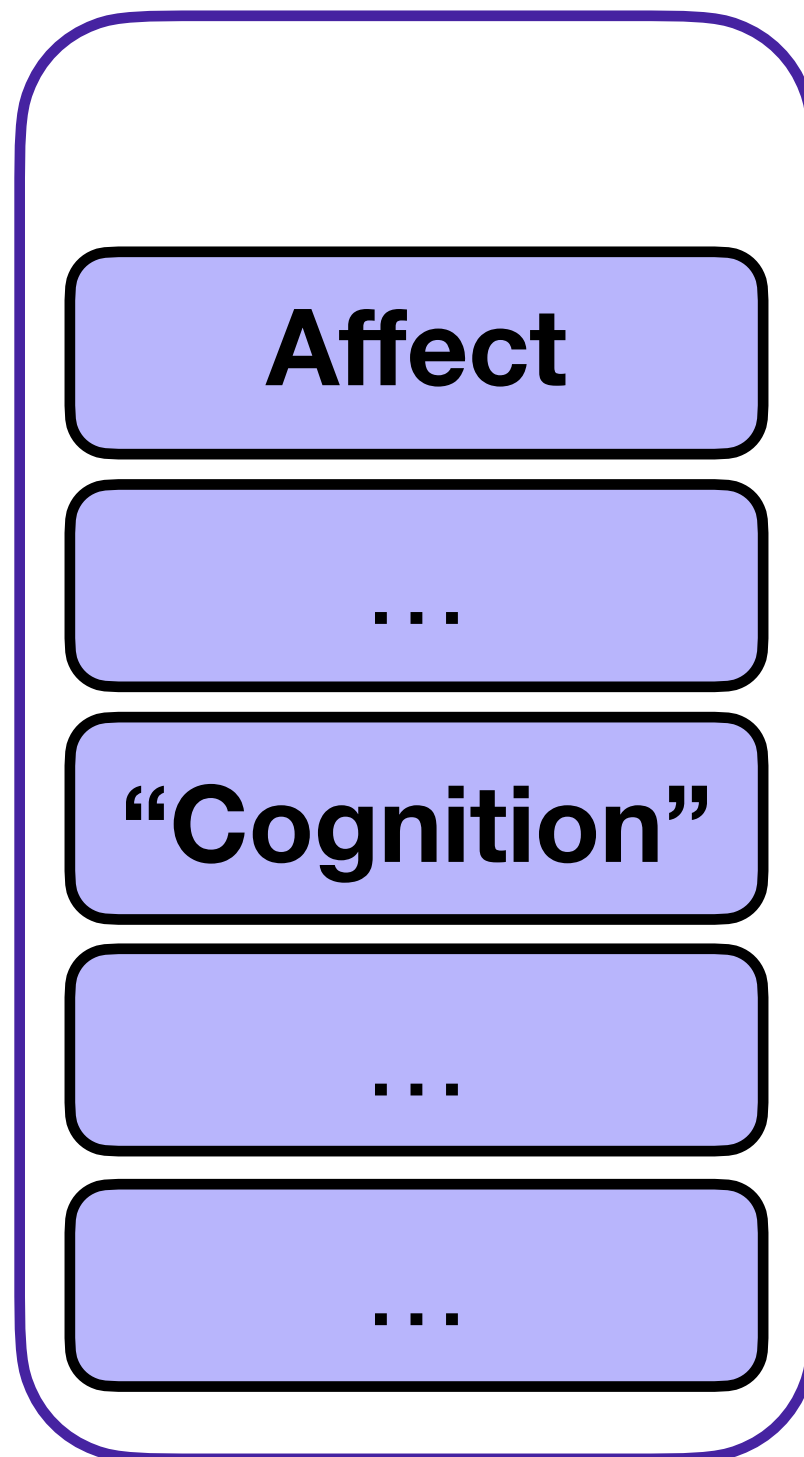
D. Experiments 1-7: Categorization

E. Handling objections

F. Experiments 8-14: Similarity

G. Discussion/Conclusion

AFH ~ prominence



Evidence for AFH

Participants respond faster + have earlier ERP responses to (affect effects):

- Affective vs. Conceptual dimensions of a single stimulus
- Emo vs. neutral stimuli

(Zajonc 1968; Fazio 2001; Costa et al 2009; Vigliocco et al 2011; Scott et al 2009; Scott et al 2012; Bayer et al 2012; Rellecke et al 2011; Palazova et al 2011, Hinojosa et al 2018, 2019, 2020; a.o. ...)

The Alternative

Object recognition
required to establish
affective response

(Lazarus 1984; Rolls 1999; Storbeck, Robinson & McCour 2006; Storbeck & Clore 2007; Calvo & Nummenmaa 2007; Batty, Cave, & Pauli, 2005; Fox et al., 2000; Horstmann & Bauland, 2006; Nummenmaa, Hyönä & Calvo 2010)

“Cognition”

...

Affect

...

...

Affect effects in language

Task-dependent & sensitive to psycholinguistic variables

(Hinojosa et al 2014, Kaltwasser et al 2013; Hinojosa et al 2019)

Appear when words describe people or are in communicative contexts

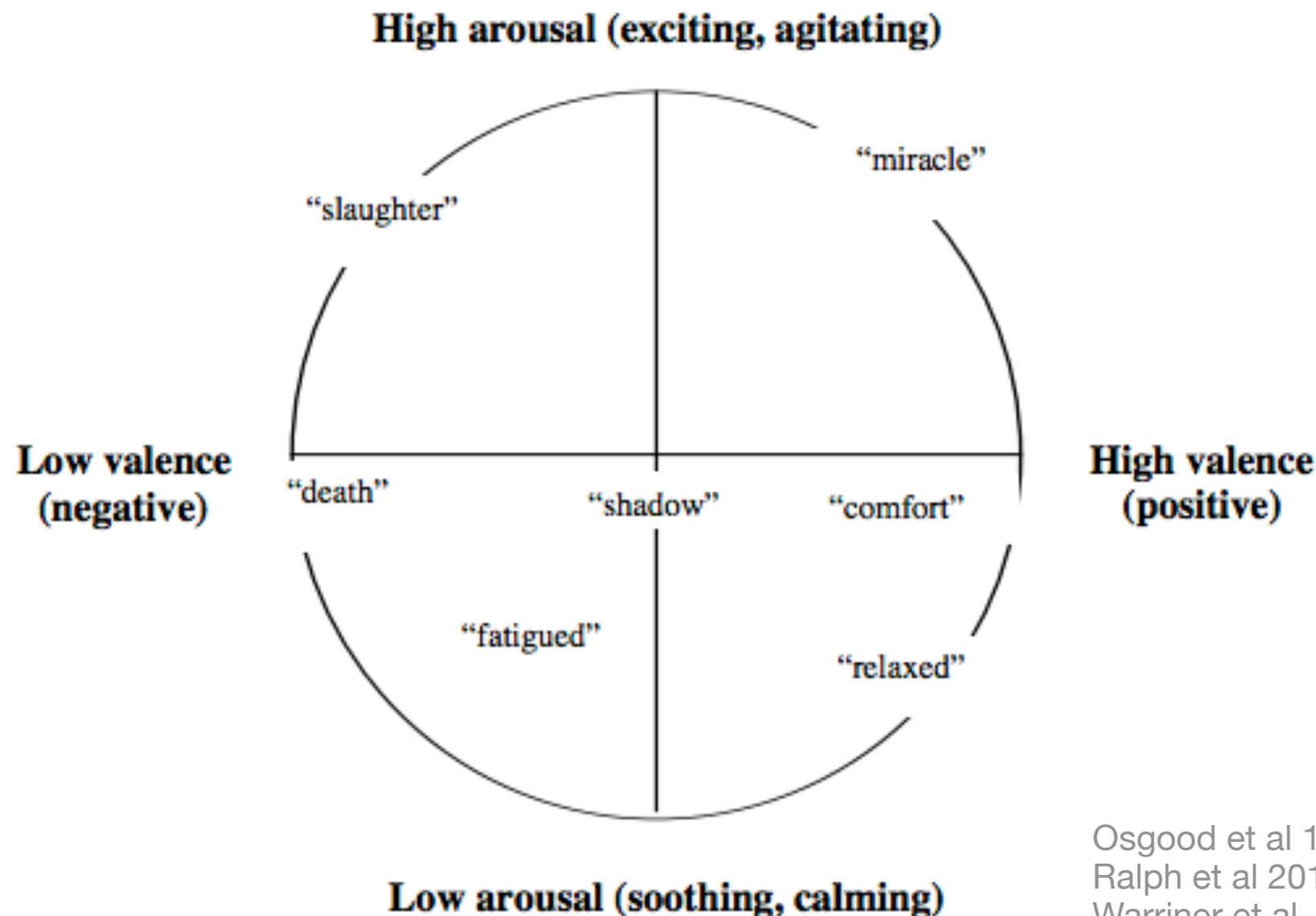
(Rohr & Abdel Rahman 2018; Rohr & Wentura 2022)

Direct listener attention to speaker intentions

(Rodrigues Ronderos & Domaneschi 2023)

Psychologists: all (most) concepts have an affective component

Figure 1. Affective experiences may be best described in two dimensions: Valence refers to how positive or negative an event is, and arousal reflects whether an event is exciting/agitating or calming/soothing. Words have been placed at locations within this space, indicating their approximate valence and arousal ratings (ratings from Bradley & Lang, 1999 [#11]).



Valence:
the positive
or negative
evaluation

Osgood et al 1957; Mehrabian et Russell 1974; Lambon Ralph et al 2017; Yap & Show 2017; Bradley & Lang 1999; Warriner et al. 2013; Mohammad 2018

Affect and the Fregean Program



Persistent puzzles to the Fregean program

Strawson (1950) “on referring”, Perry (1979) “Essential indexical”, Lewis (1979) “attitudes de dicto and de se”, Davidson (1986) “derangement of epitaphs”, Lewis (1996) “elusive knowledge”, Kaplan (1999) on “oops and ouch” & Kratzer’s discussion of it...everything in the pragmatic wastebasket

From ~2005 on explosion of work on expressives, slurs, evaluative, and other non-truth-conditional aspects of meaning

Potts (2006), McCready 2014, Gutzmann (2015), Cepollaro (2020), Jeshion (2021)...Stojanovic (2007), Stojanovic (2012)

Current Project

Affect

Concept. 1

Concept. 2

Concept. 3

Concept. 4

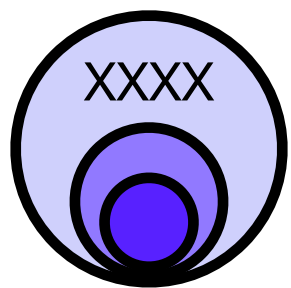
Valence x conceptual foils

Exps 1-4: Is valence categorization faster?

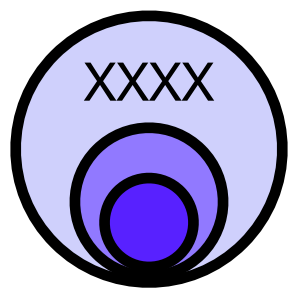
Exps 5-9: Is valence more important for similarity?

Conceptual foils

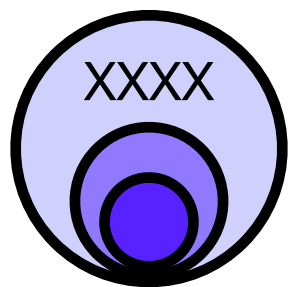
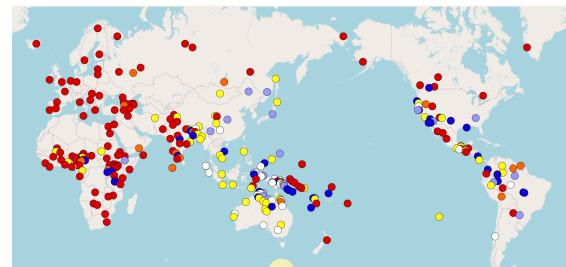
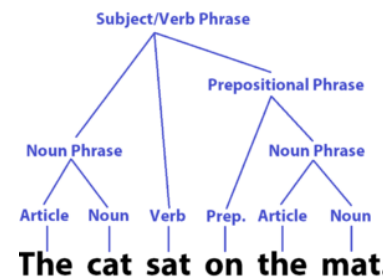
1. Concrete vs. abstract
(nouns, verbs, adjs)



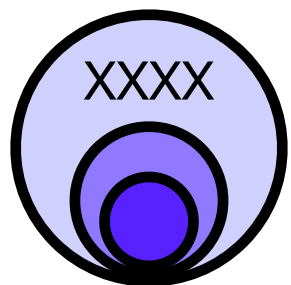
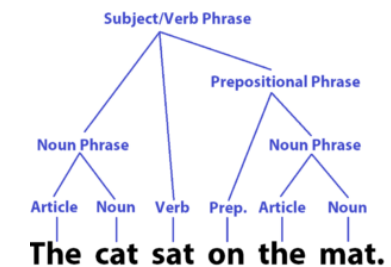
2. Physical vs. psych./social
(verbs, adjs)



3. Animate vs. inanimate
(nouns)



4. Syntax
(noun vs. verb)



Roadmap

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B. Affect-First Hypothesis

C. Conceptual Dimensions

D. Experiments 1-7: Categorization

E. Handling objections

F. Experiments 8-14: Similarity

G. Discussion/Conclusion

GENOCIDE

F

positive

J

negative

GENOCIDE

F

abstract

J

concrete

Speeded categorization

2 x 1 within-subjects

blocked design (counterbalanced)

- Task (**Valence**, **Conceptual**)
- Items presented 3x per block (randomized)
- DM: key board press F or J (counterbalanced)

7 exps

270 Native American English speakers on
Prolific, £9/hr, ~15 min median time

Stimuli

40 words in each exp, within lexical category

10 per 4 possible feature combination

Feature values came from corpus or pre-test norming

Stimuli

- **Valence**: Warriner et al 2013
- **Conc. vs. Abs.**: Brysbaert et al 2013
- **Phys. vs. Psych./Soc.**: pre-test norming
- **Animacy**: Wordnet synsets with 'living organism' or 'alive' as superset (+intuitions)
- **Syntax**: unambiguous nouns vs. verbs

RT Analysis

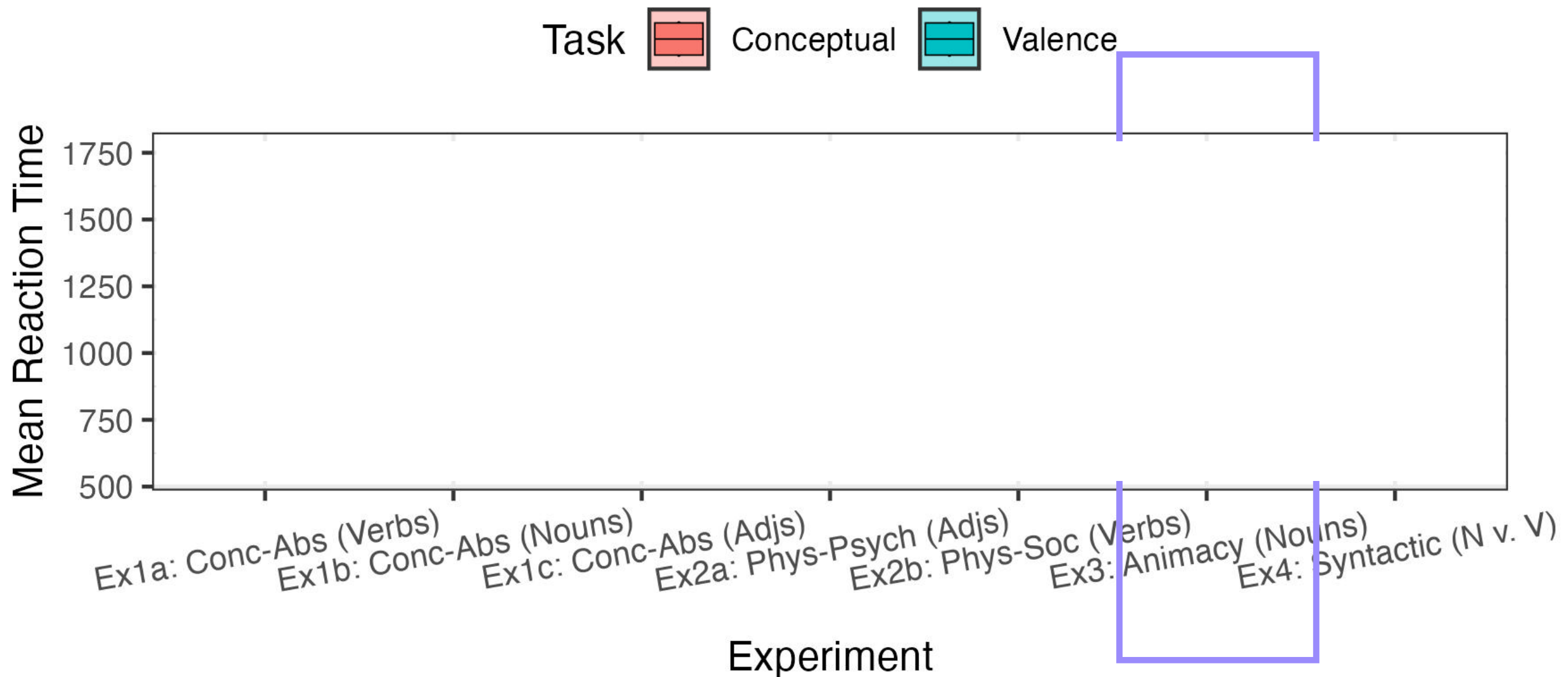
- Inaccurate participants removed ($\text{Acc} < 75\%$)
- Only Accurate trials included
- Excluded trials where $\text{LogRT} > 3 \times \text{IQR}$

Range of data excluded across exps: ~10-15%

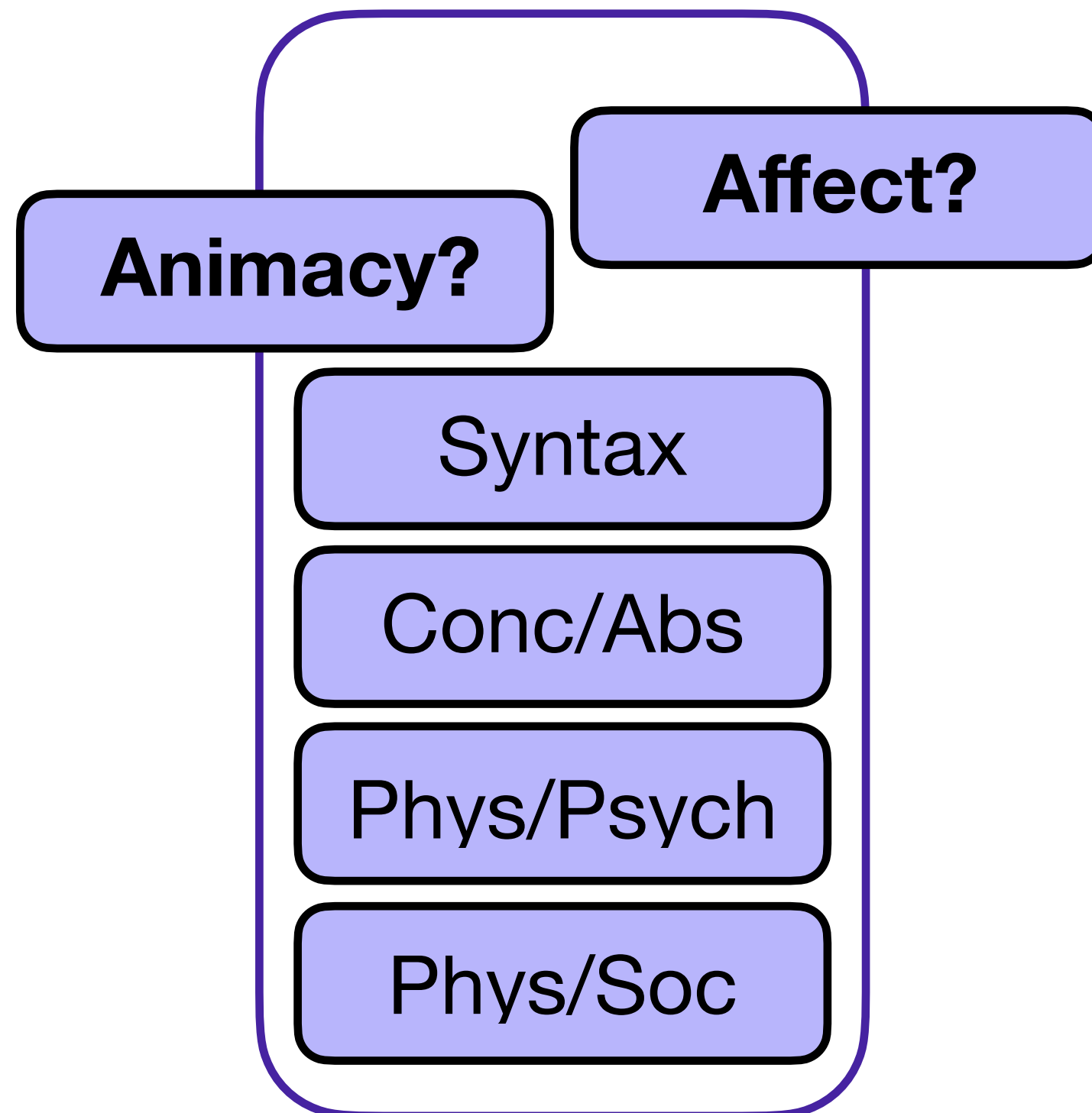
Model:

`lmer( LogRT ~ Task + ( 1 + Task | ID ) + ( 1 + Task | Word ) )`

Results: Categorization

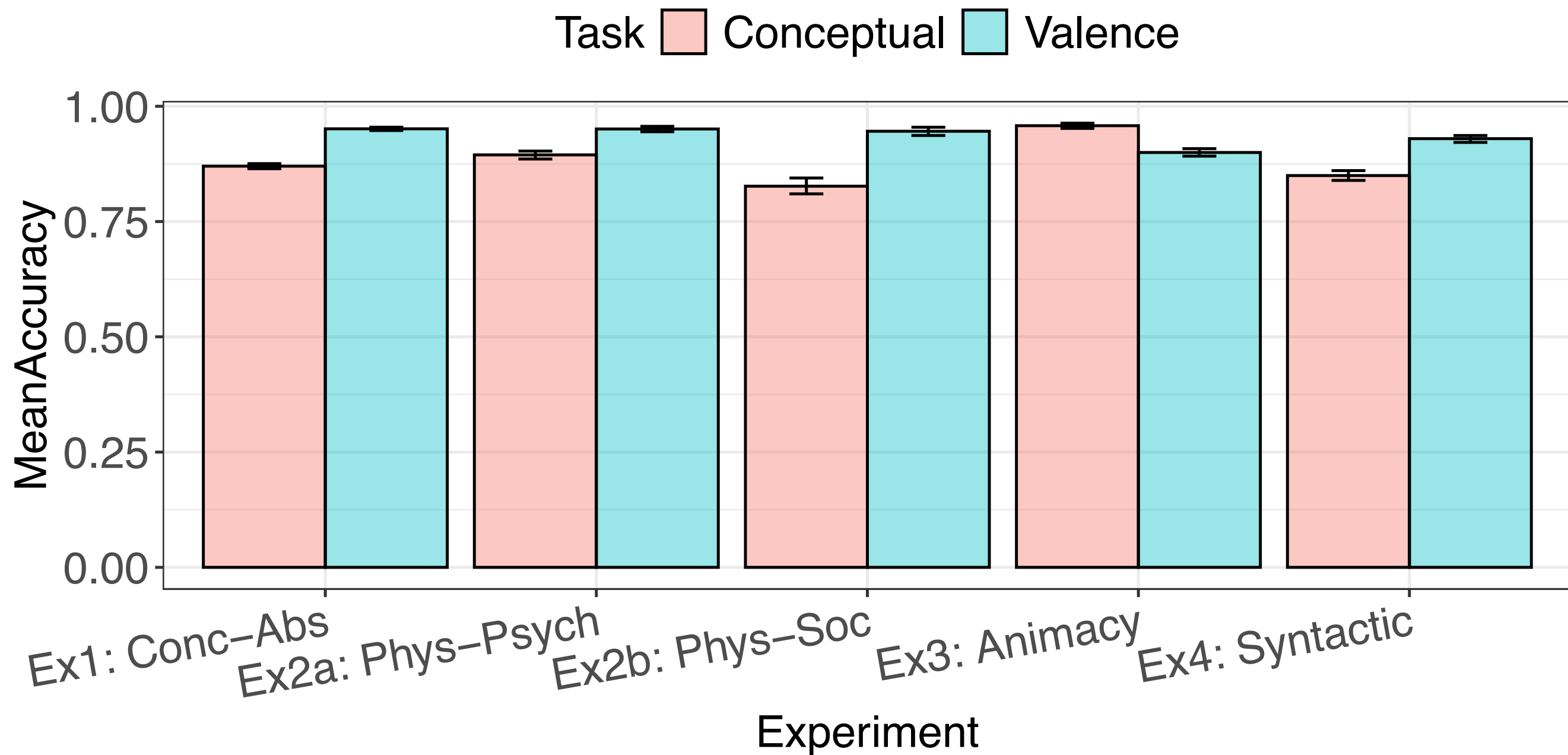


Affect ? Animacy



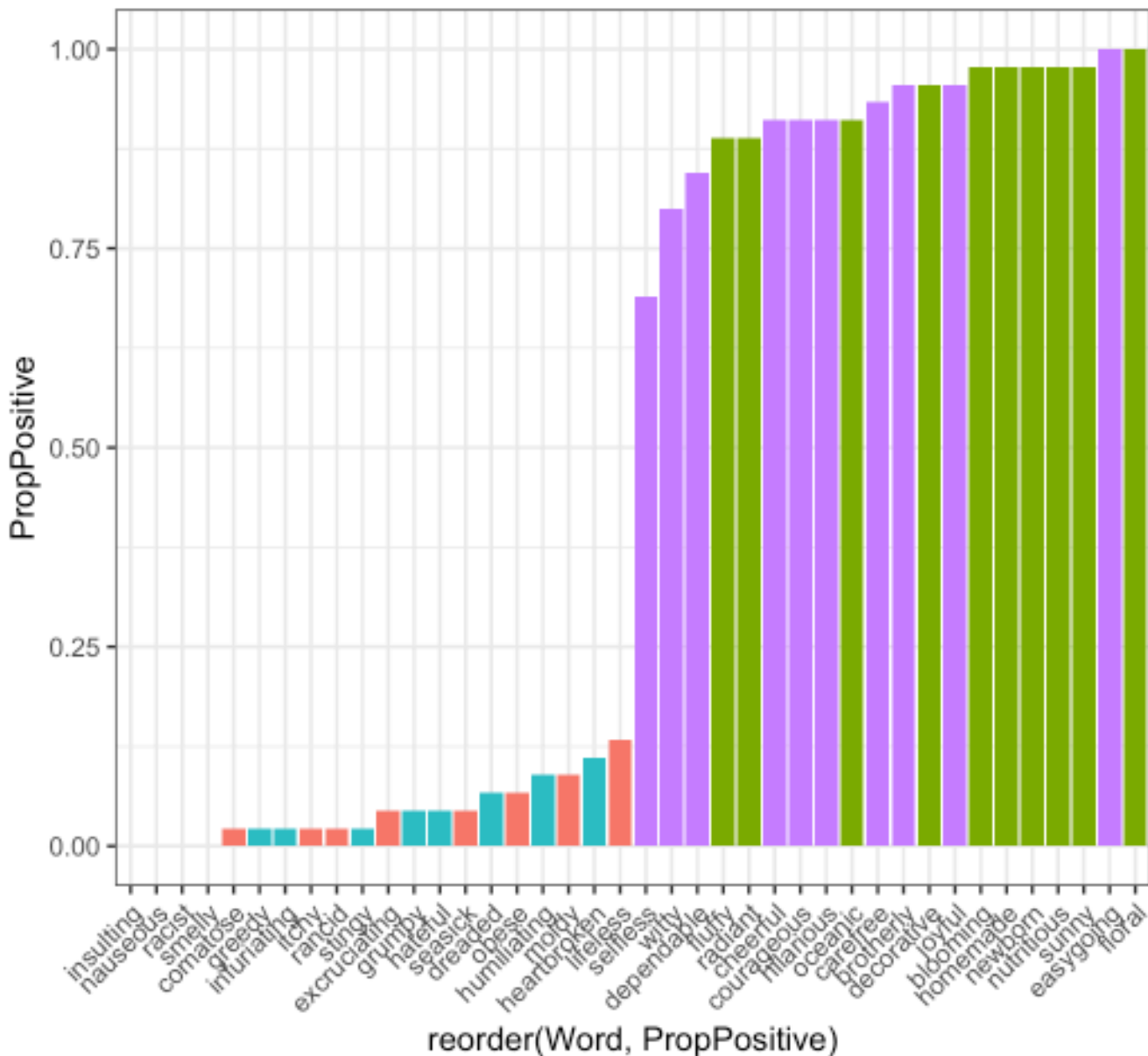
**Not just uncertainty about
conceptual meaning**

Total Accuracy

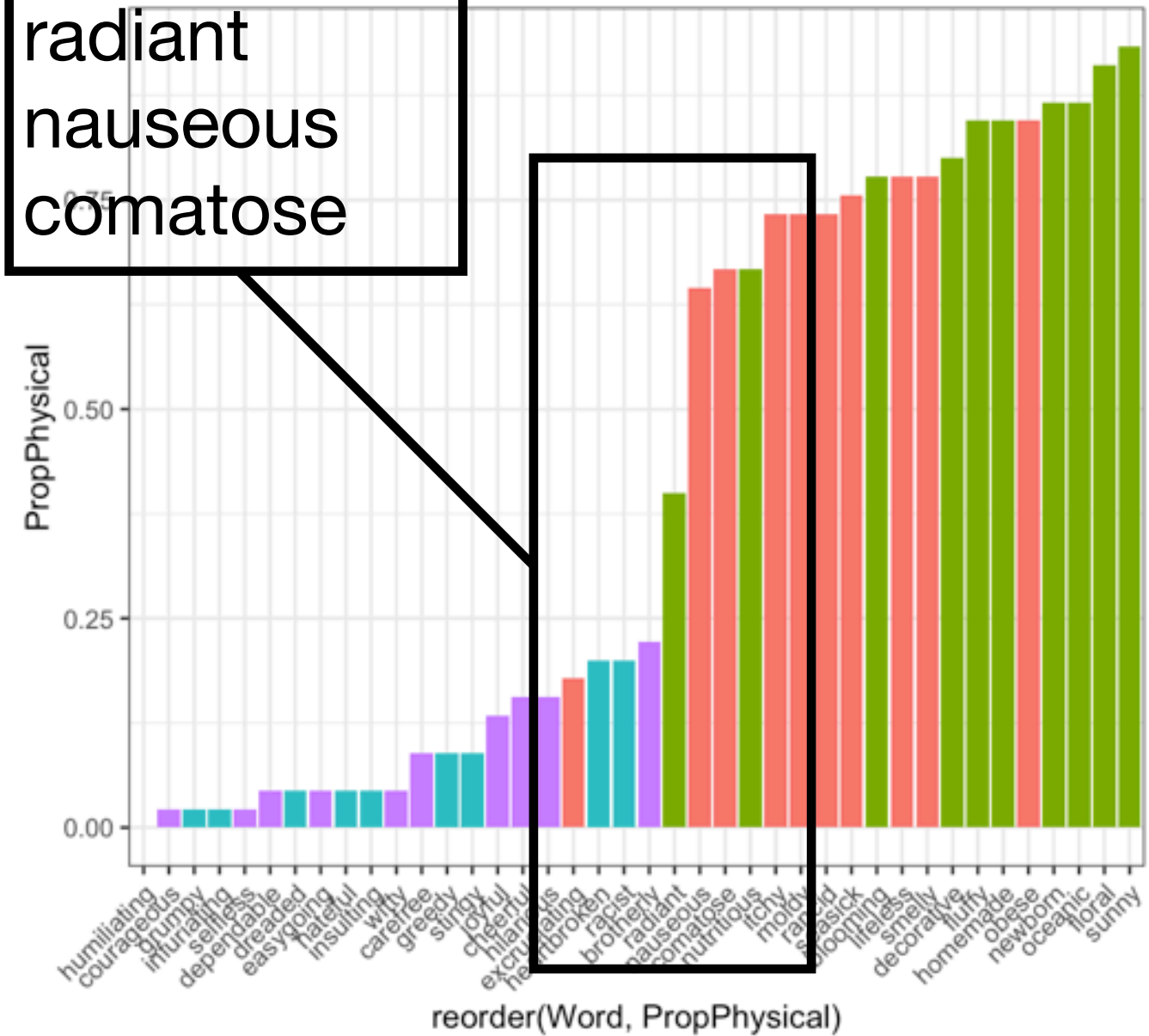


Responses 2b

Feature Combo

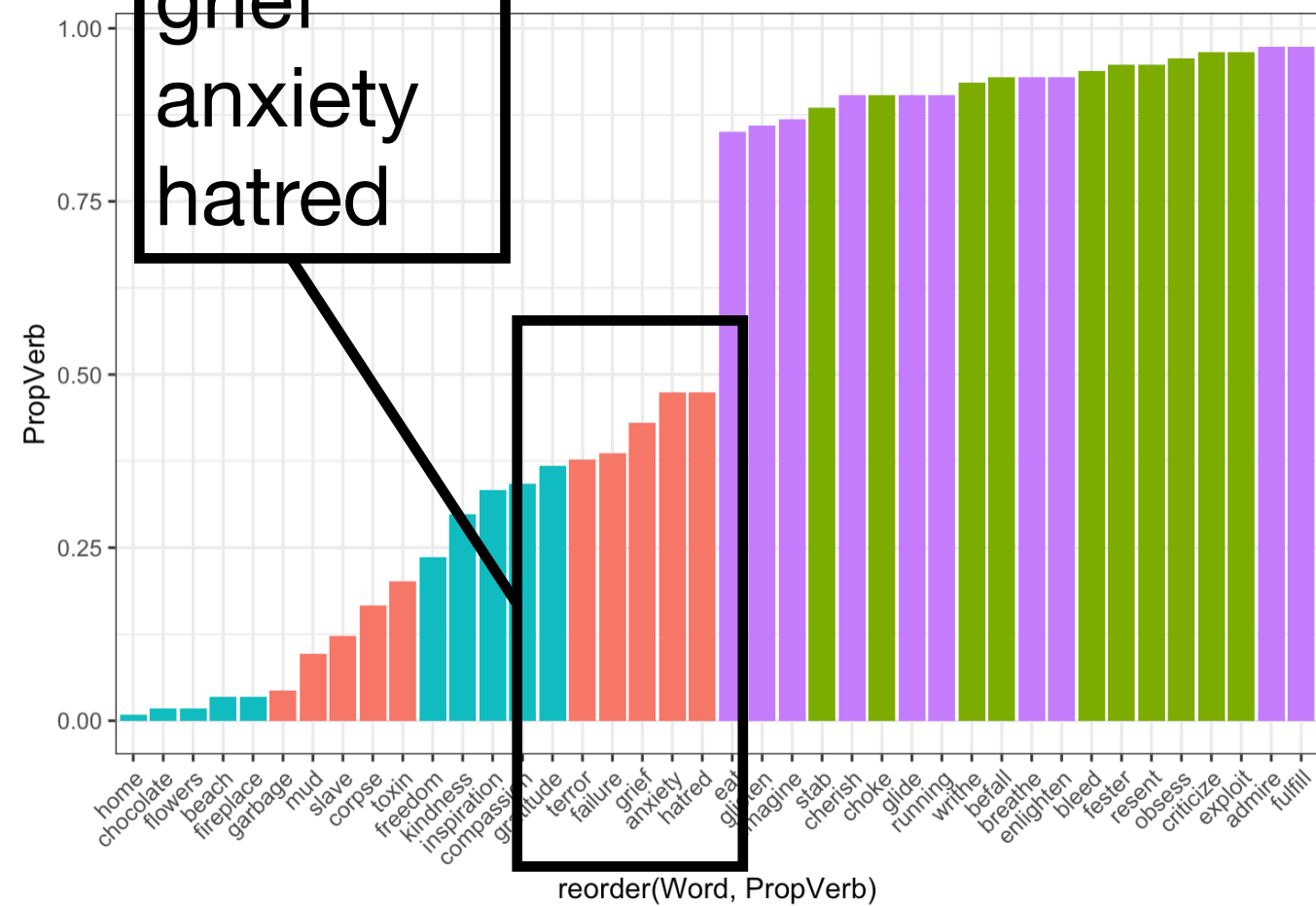
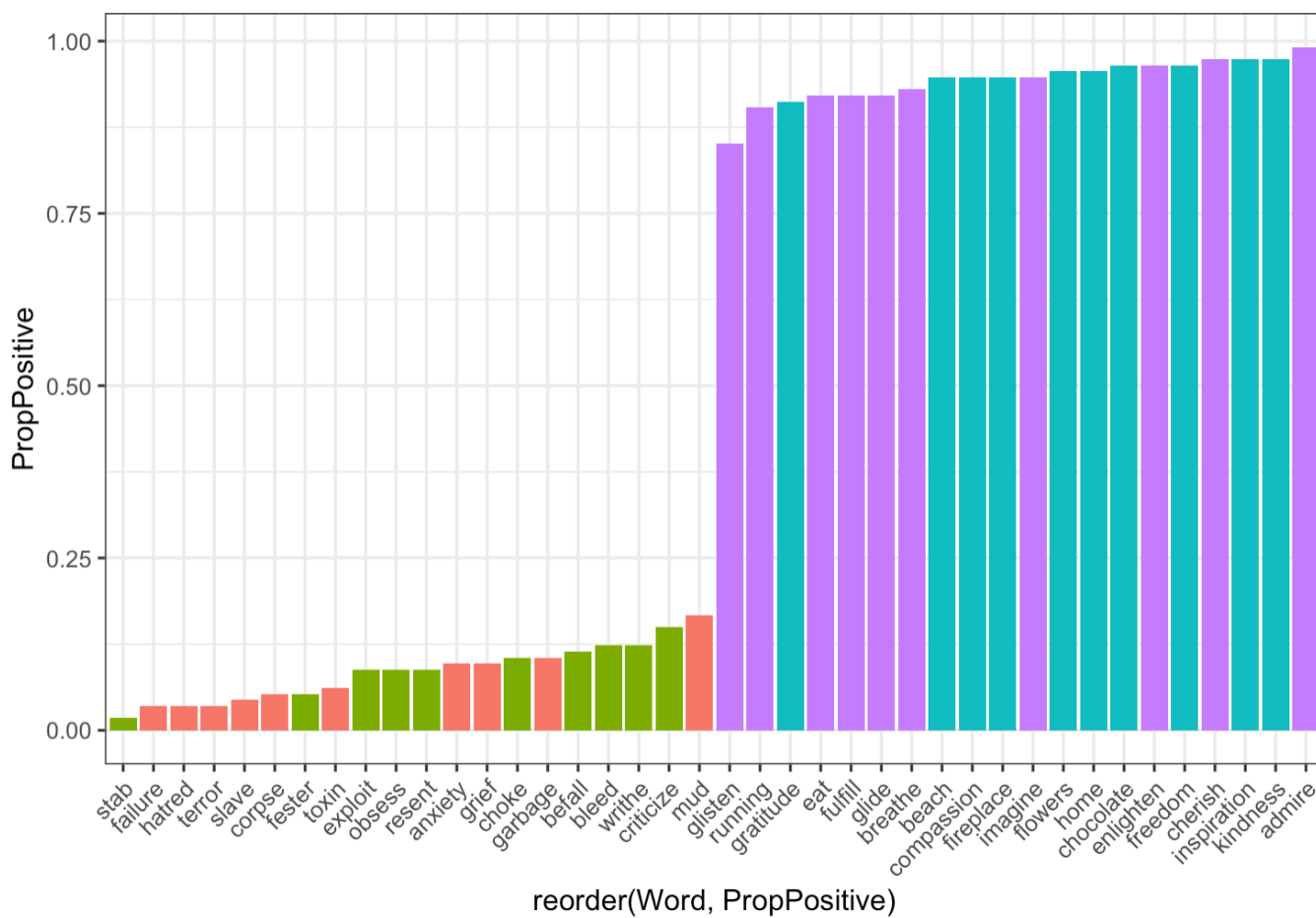
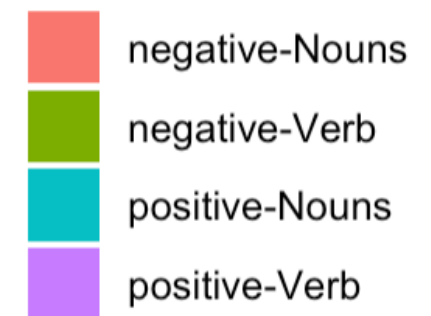


excruciating
radiant
nauseous
comatose



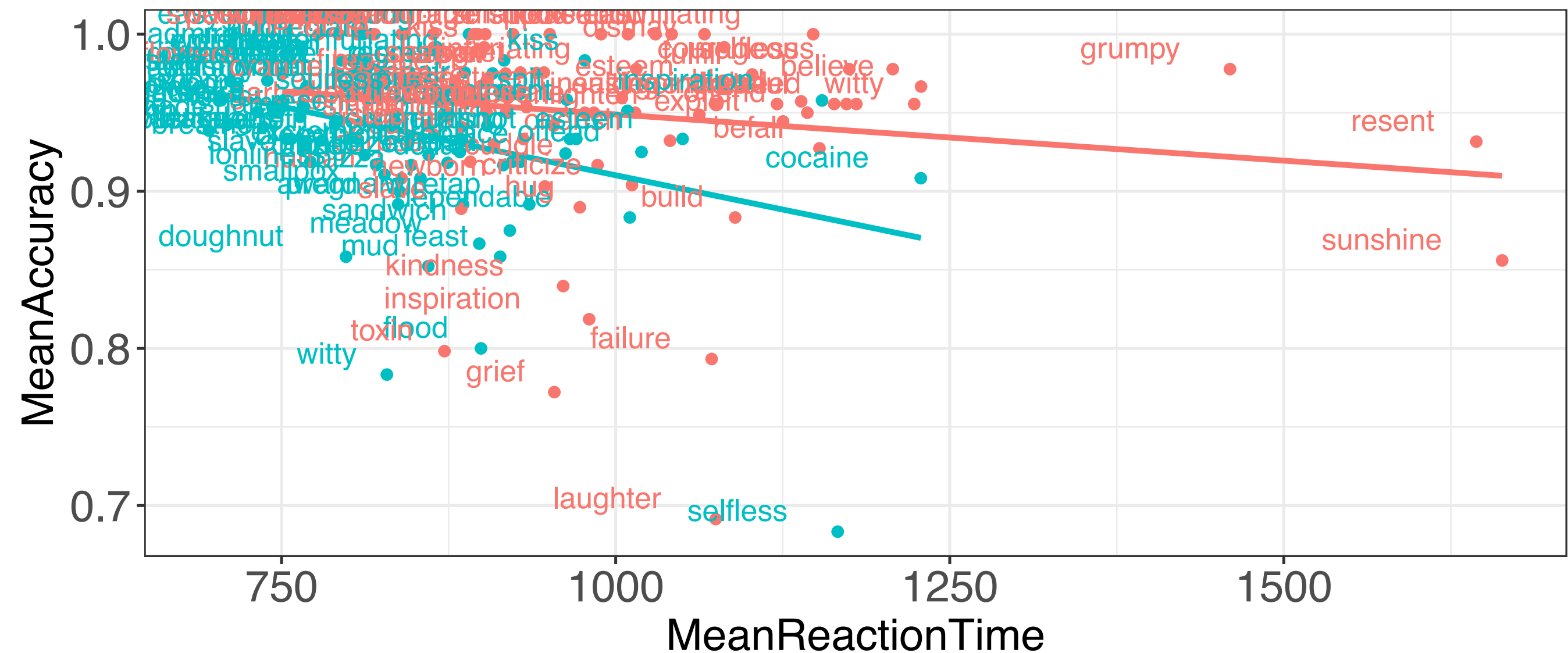
Responses Exp4

Feature Combo

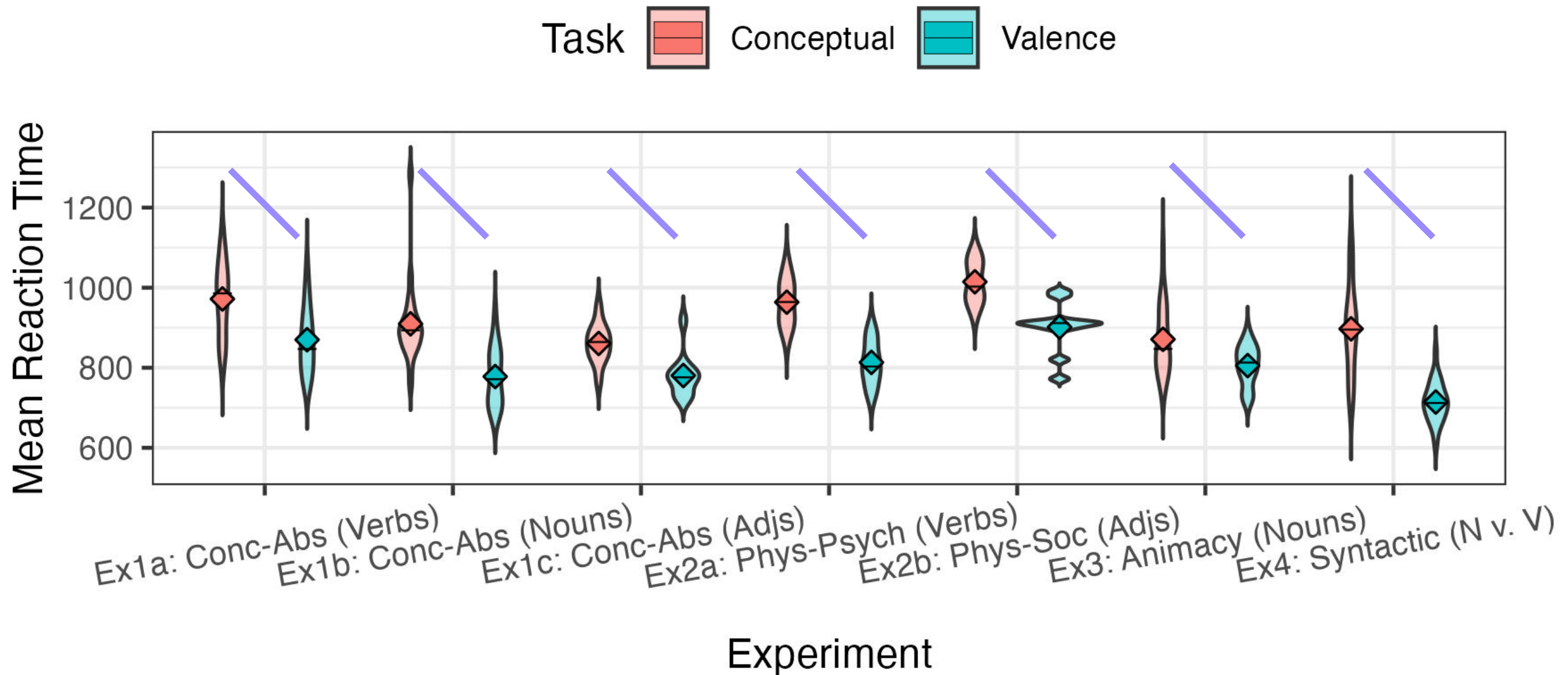


ACC_{Conceptual} ≥ ACC_{Valence}

Task —●— Conceptual —●— Valence



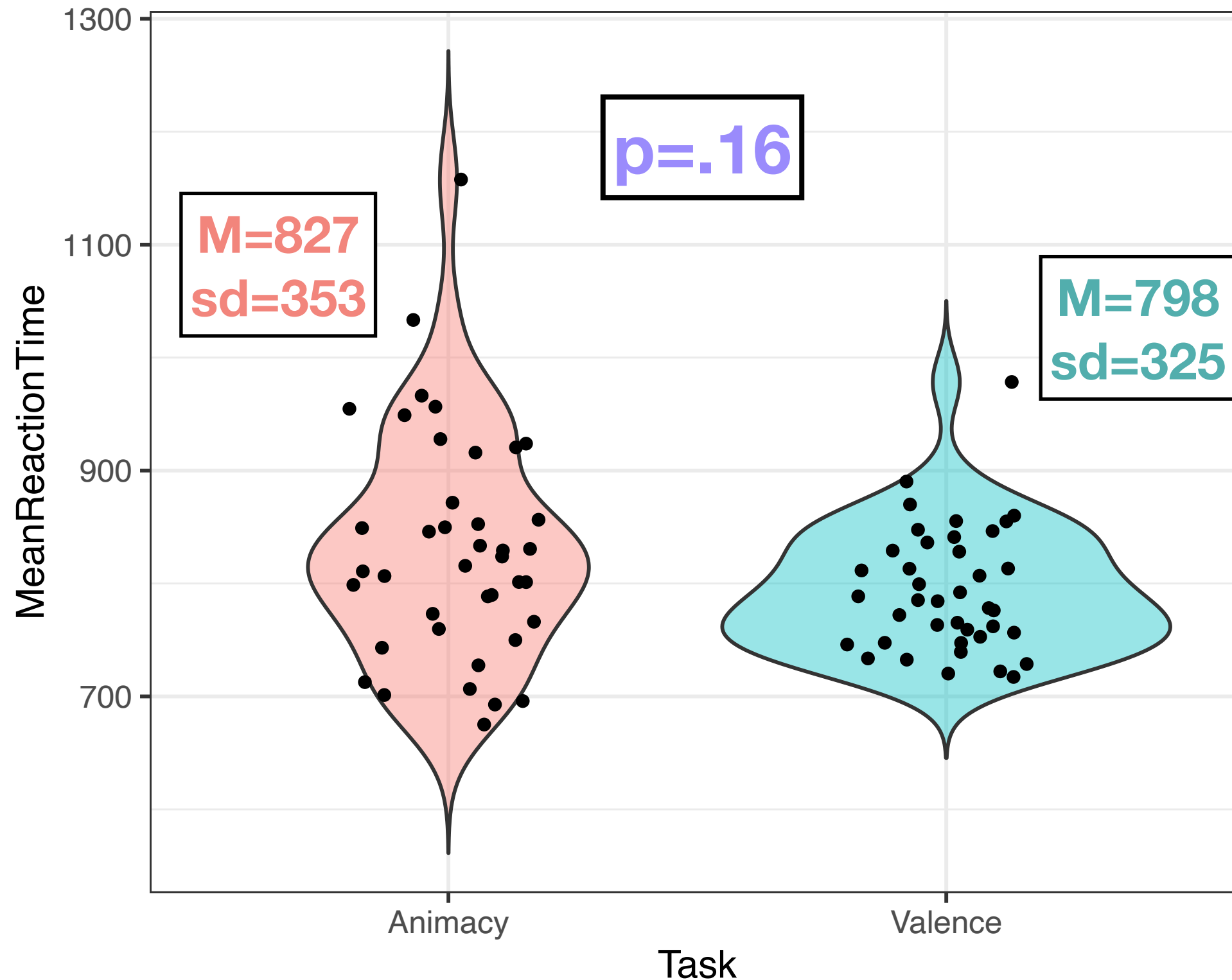
Results: Most Accurate



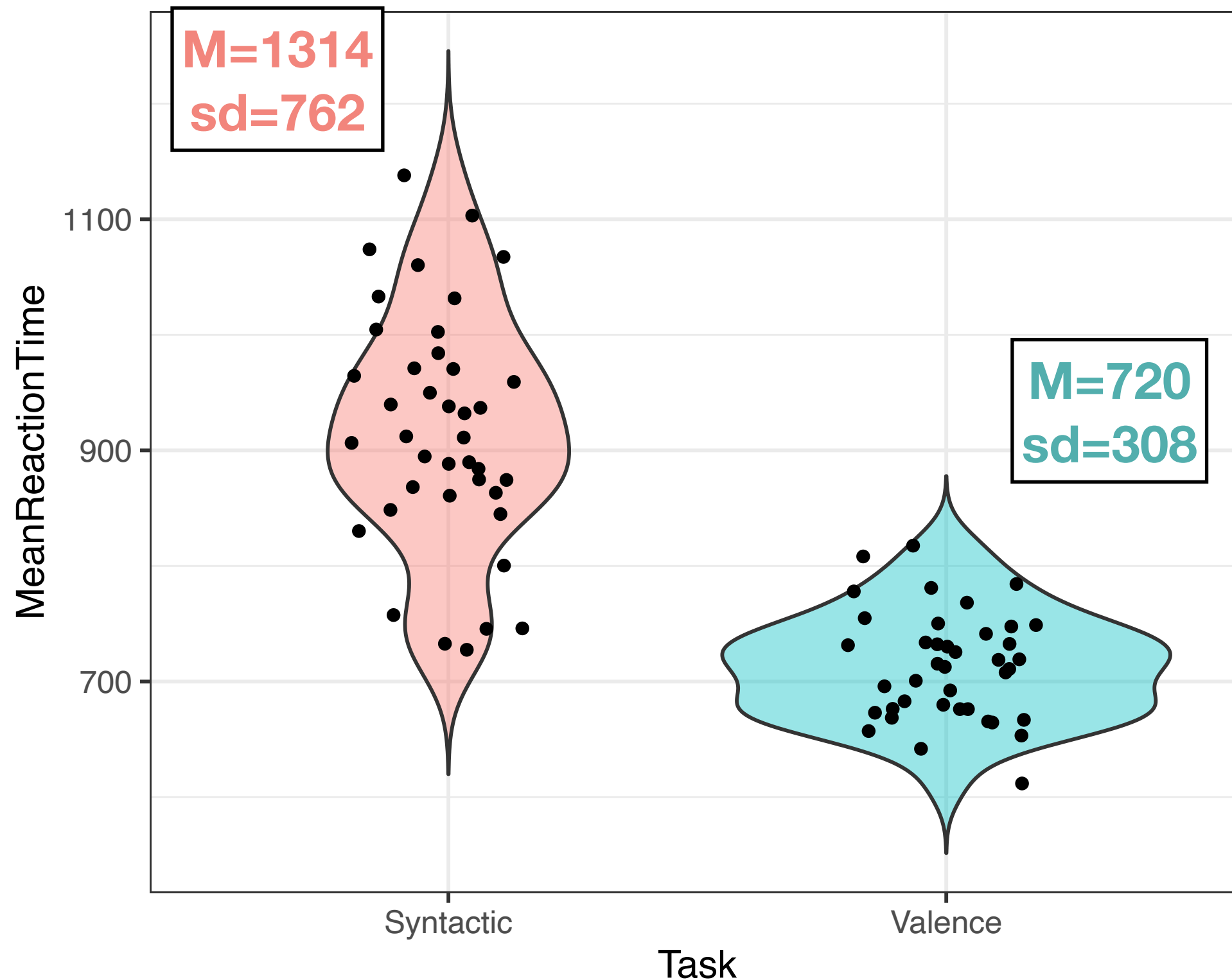
Not just uncertainty about
conceptual meaning

Not just Habit

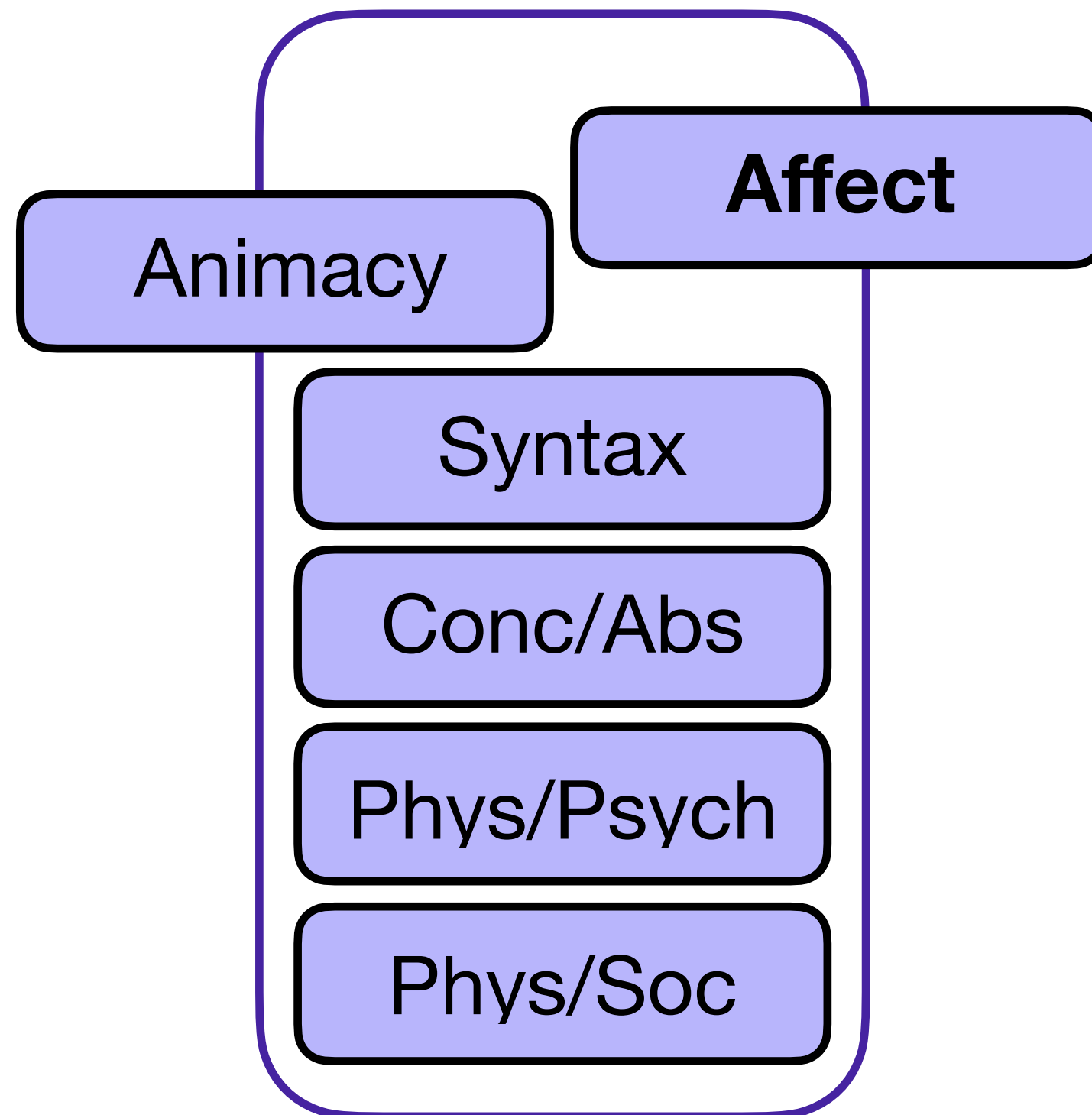
Why isn't there a greater advantage over Animacy?



Why isn't Syntax performing better?



Animacy vs. Affect



Is affective information more prominent for determining word similarity?

progress

push
pull

How similar is the word in red to the word in green?

completely
dissimilar



completely
similar

Next

Similarity judgements

Implicit measure of word meaning

(Osgood et al 1957; Degen 2015; De Deyne, Perfors & Navarro 2016, a.o.)

4 x 1: Feature Match

DM: slider from 0 (dissimilar) to 100 (similar)

MaxMismatch, DissimilarControl (Antonyms)

MaxMatch, SimilarControl (Synonyms)

**Matching
ValenceOnly**

vs.

**Matching
ConceptualOnly**

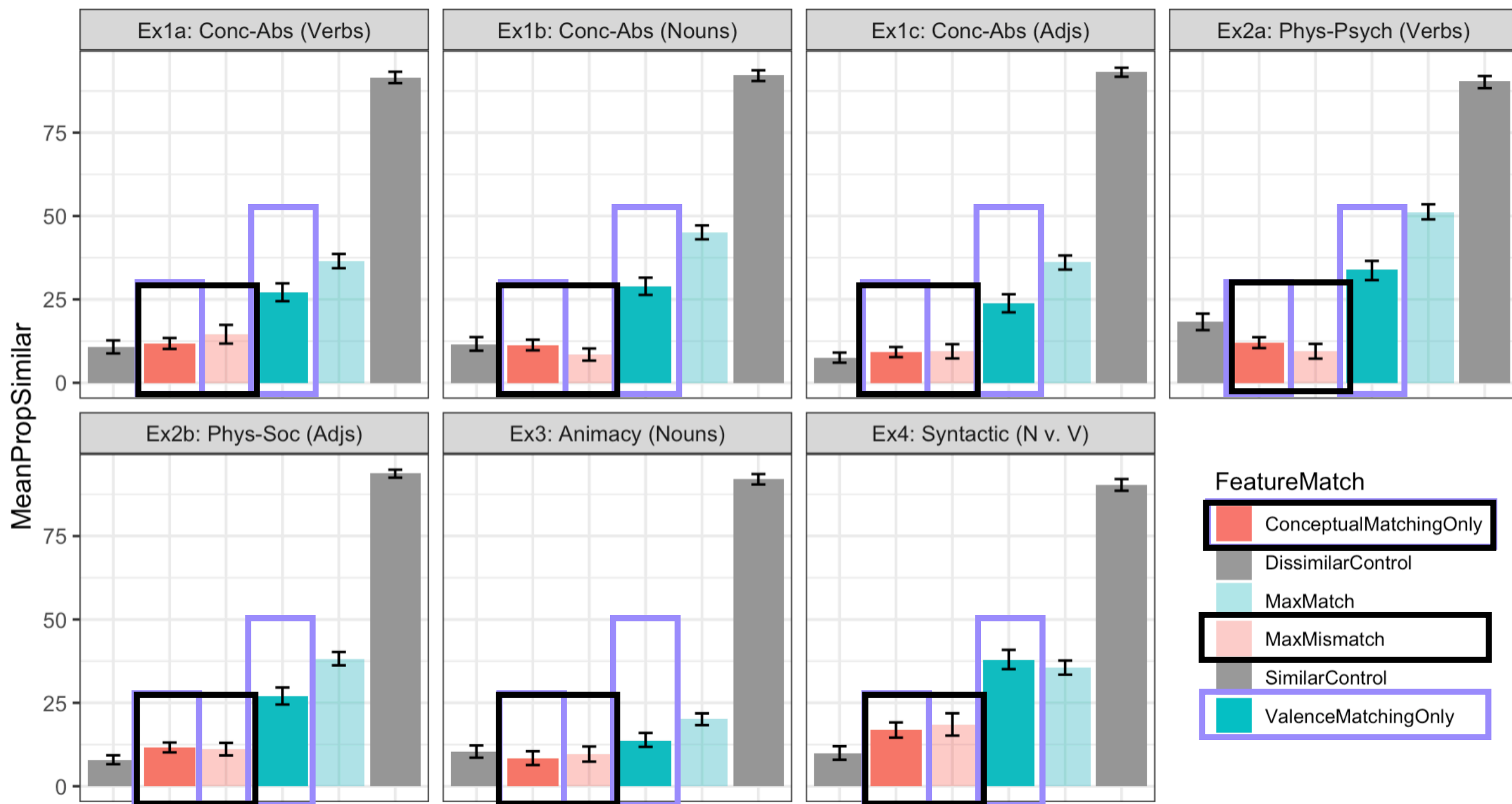
Stimuli

Words from categorization experiments

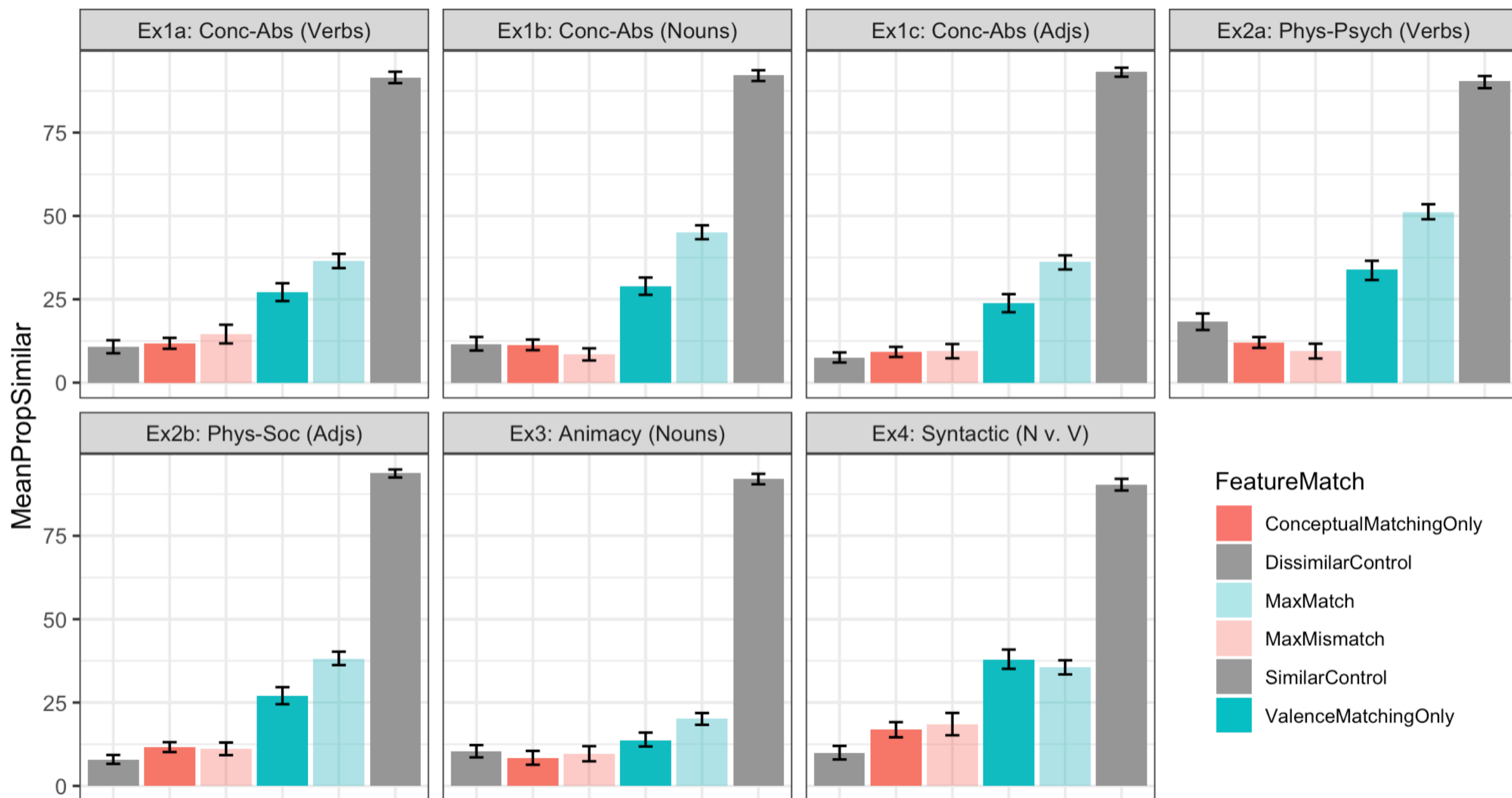
All possible combinations

- Unordered
- Same-word pairs removed
- Randomly sampled from each FeatureMatch
- 60 items total

Results (n=350)



Results (n=350)



Support for AFH

Valence faster than most Conceptual categories except Animacy

Persistent effect: found in majority of words tested

Valence, not Conceptual, features determine word meaning (similarity)

Conclusion

Affective information is **prominent** in both processing and use

Crucial component of word meaning

Affect

Animacy

Syntax

Conc/Abs

Phys/Psych

Phys/Soc

Thank you!

**Projet SU - Emergence "REV" OP24-126
Valence Asymmetries, ERC Advanced Grant,
grant n° 101142133**

Attendees of EPITHETS-STAHN 2025, New — Experimental —
perspectives on Valence 2025



Experiment 1a-c:

concrete / abstract

Exp 1a-1b (n=90)

Using Corpus norms for Valence (Warriner et al 2013)
and Concreteness (Brysbaert et al 2013)

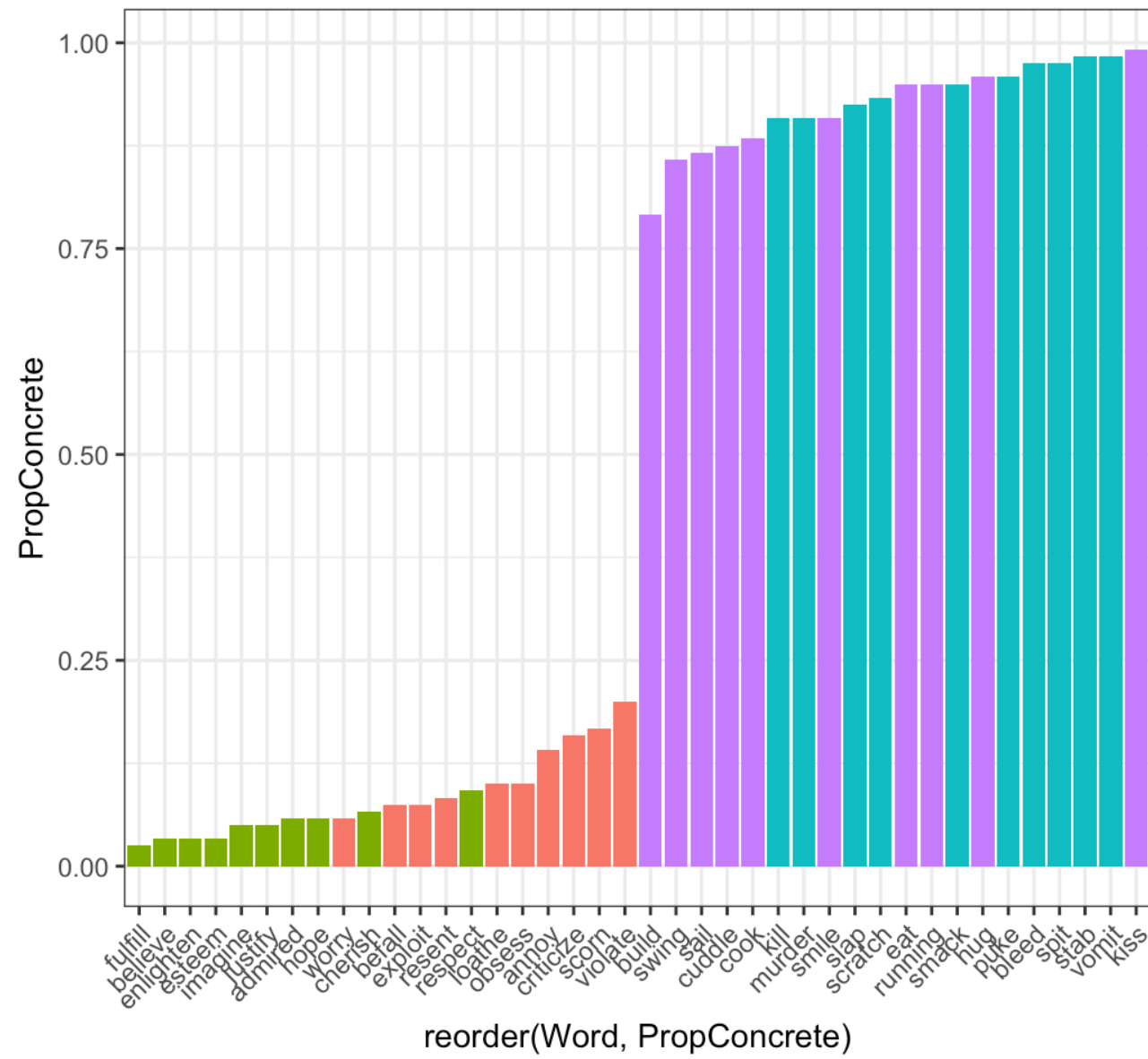
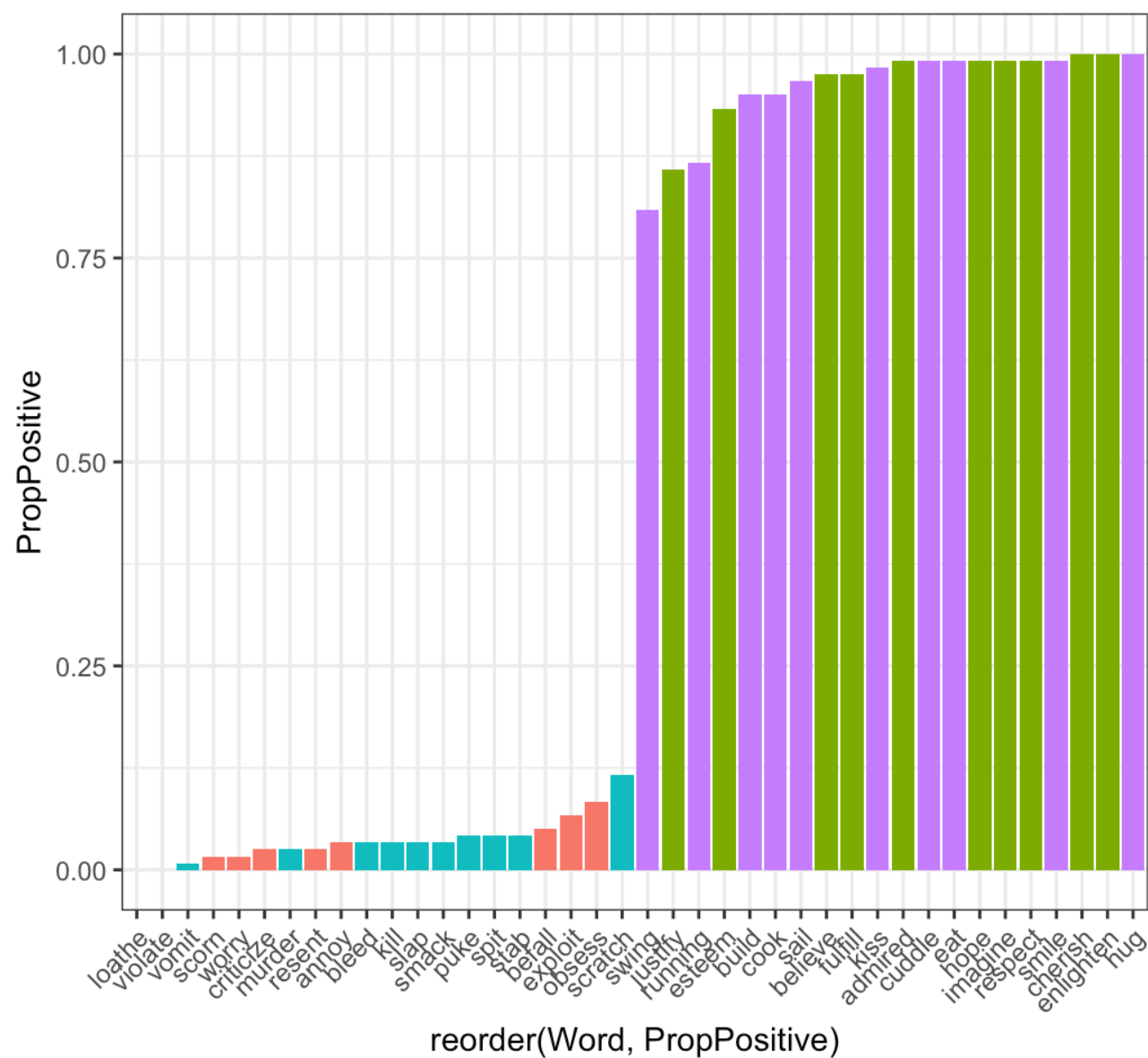
High Frequency, and within +/-2sd of mean Word Length, and overly valanced morphosyntax

40 words per, 10 from each combo of Conc+Val

- Maximizing mean rating for concrete/positive
- Minimizing mean rating for abstract/negative
- Minimizing rating SD

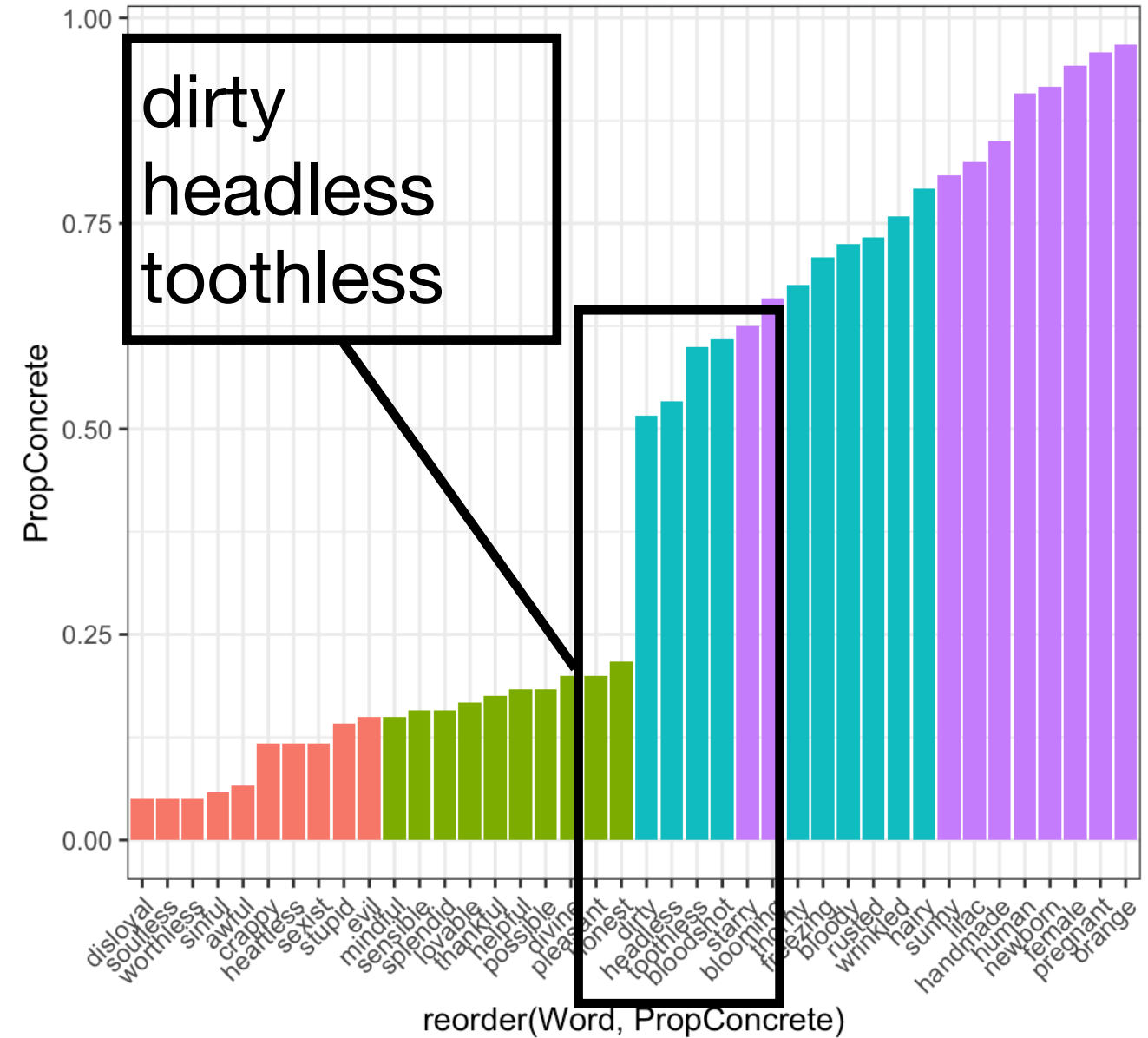
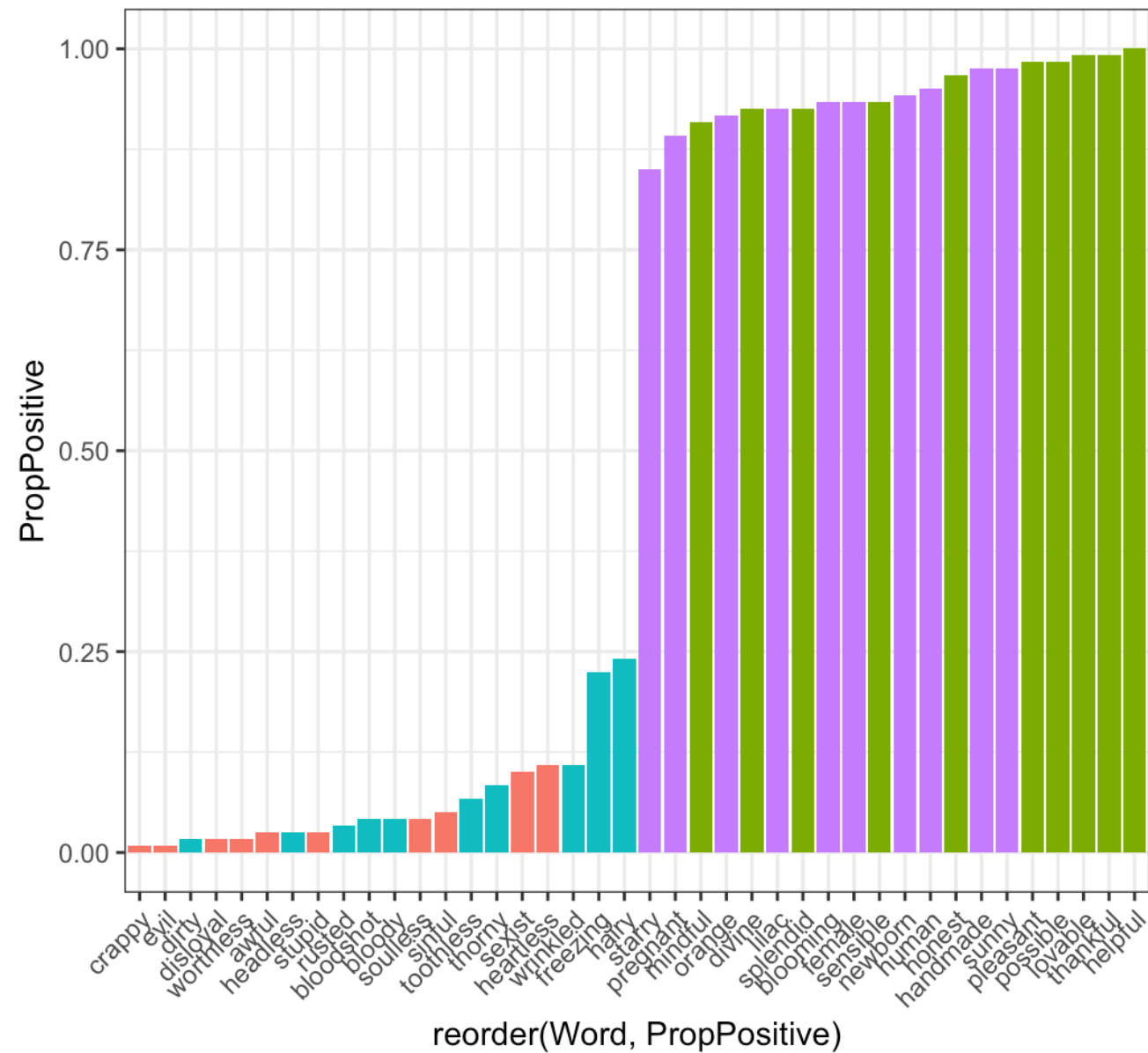
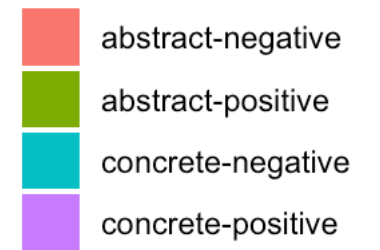
Responses - Verbs

Feature Combo

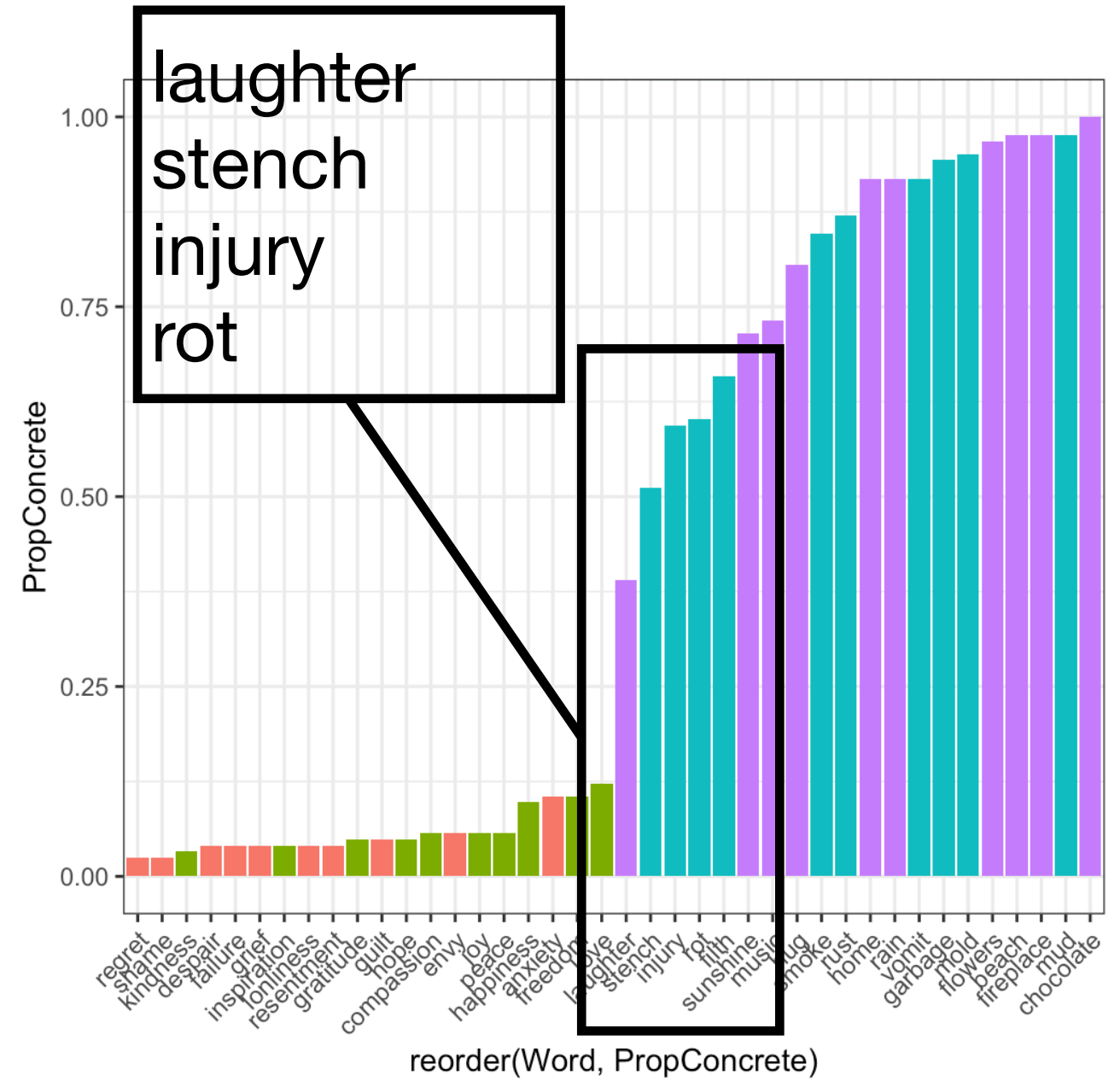
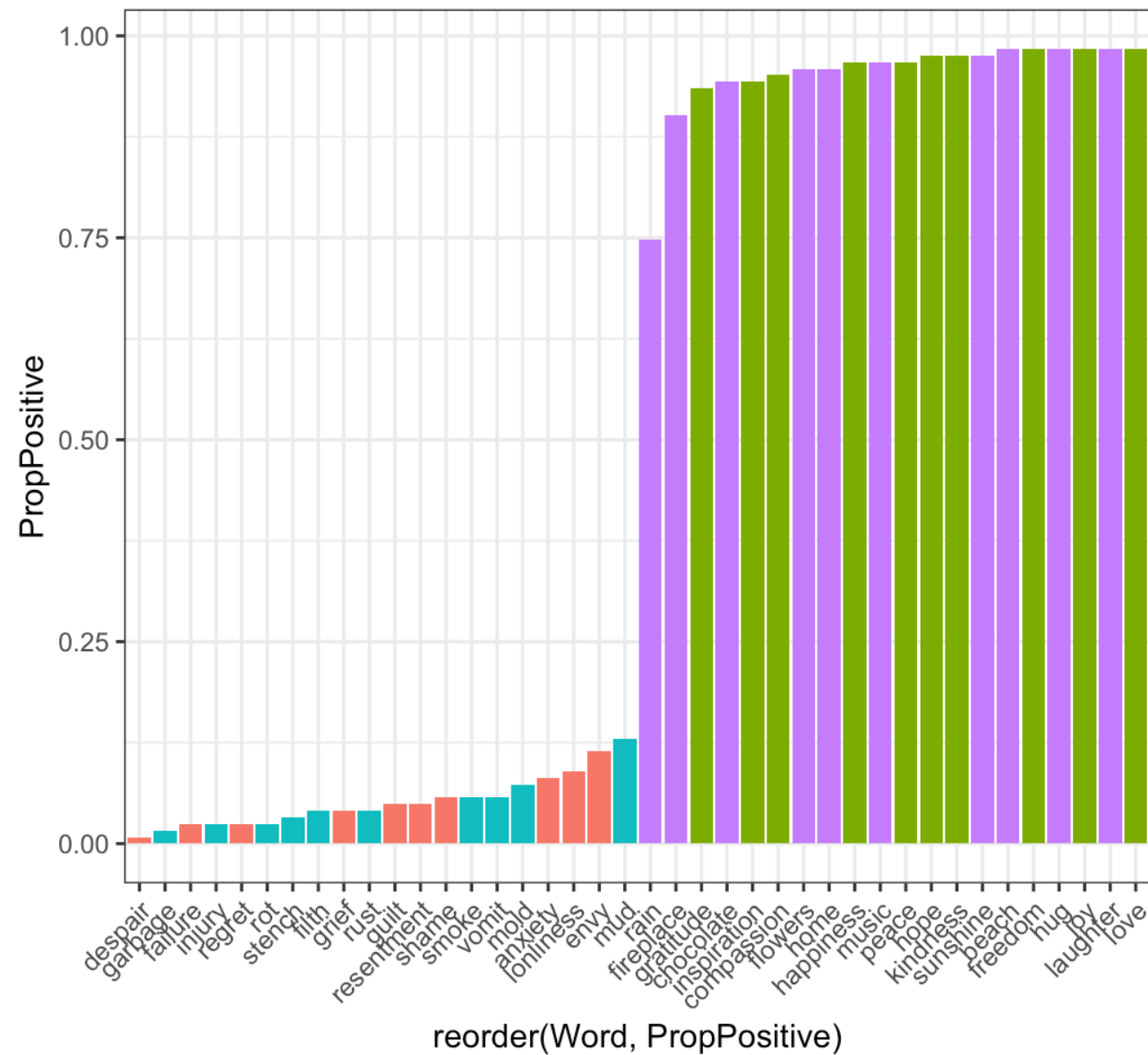
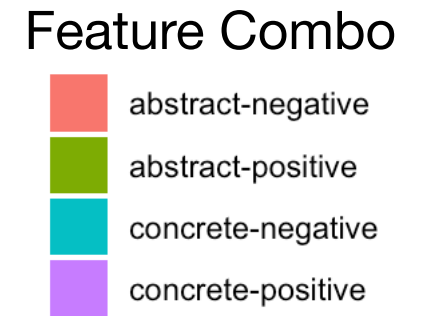


Responses - Adjs

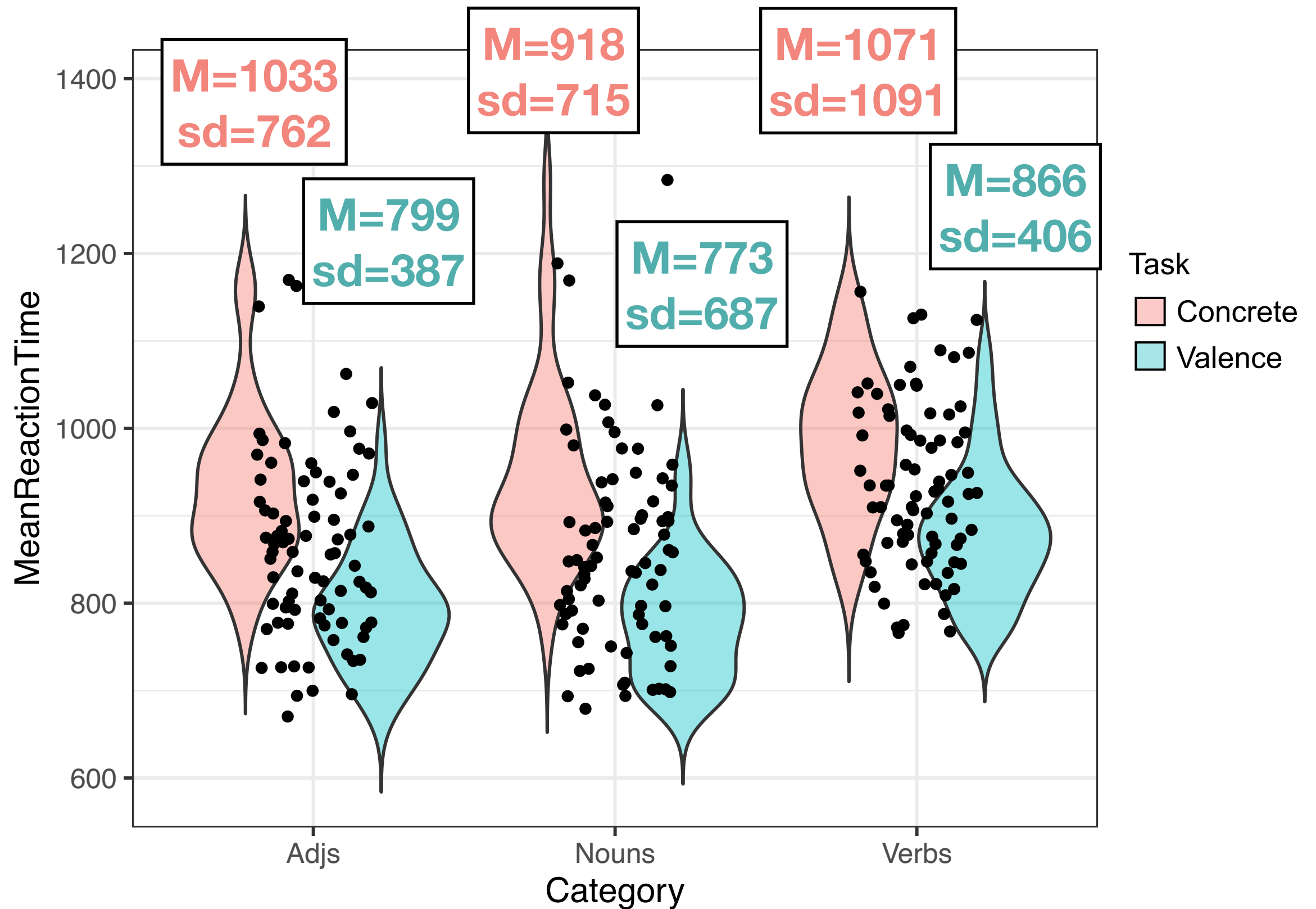
Feature Combo



Responses - Nouns



Experiment 1 Results



Adjs



Nouns



Verbs



Experiment 2a-b: physical vs psychological (verbs)/ social (adjs)

Exp 2 Stimuli

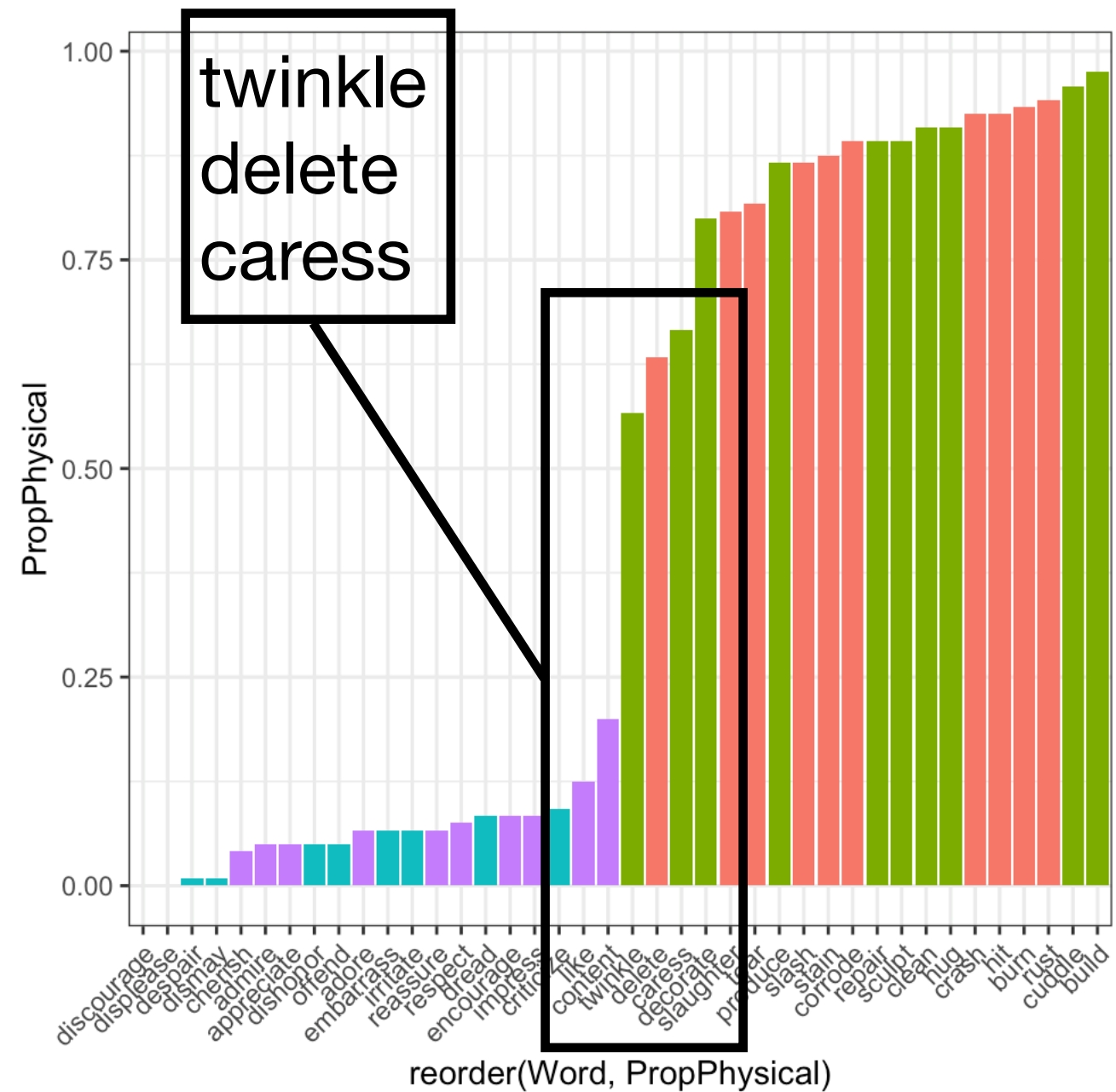
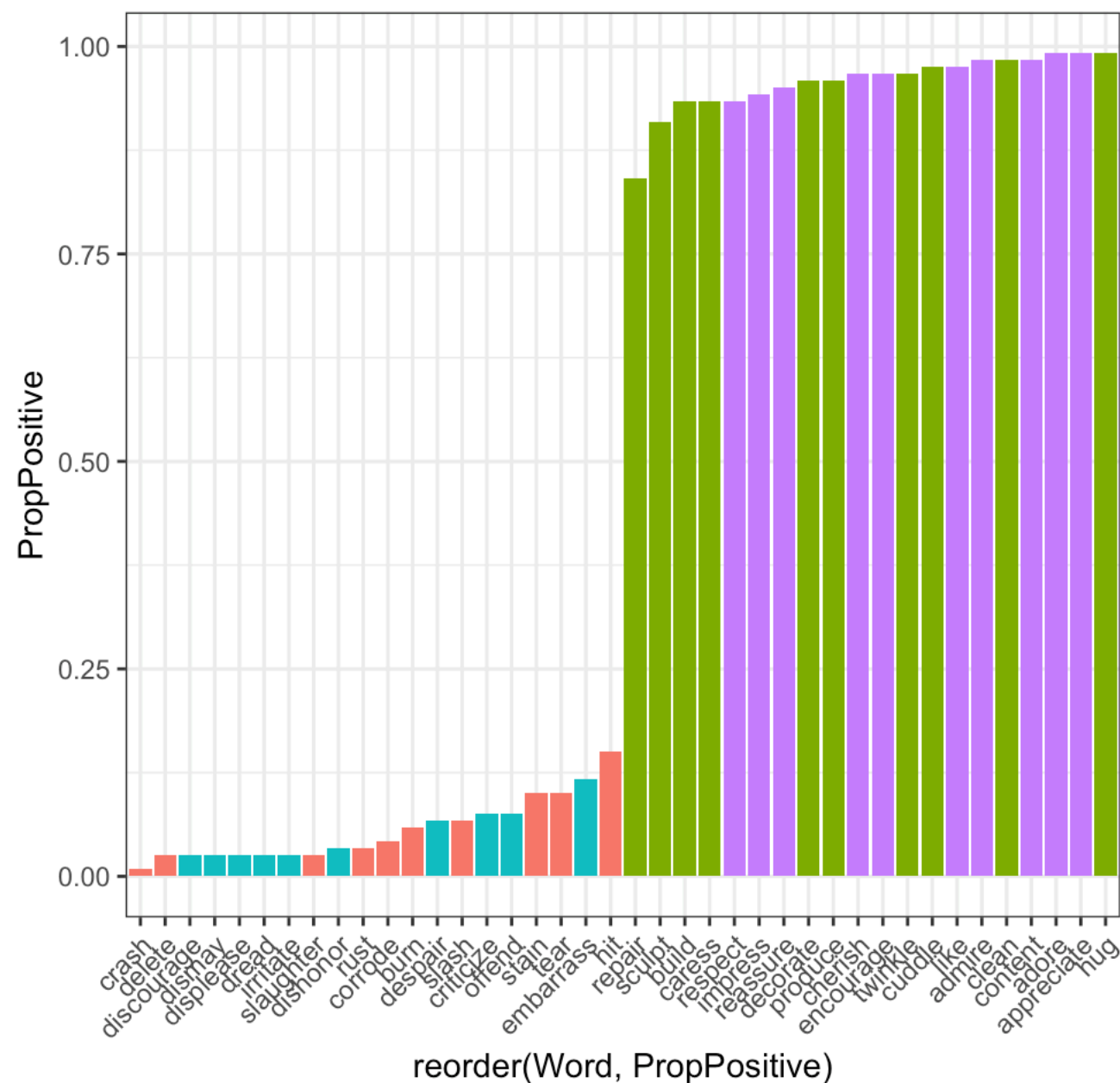
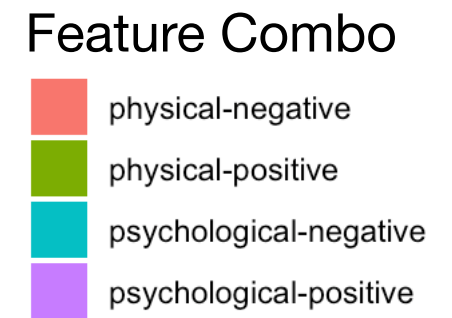
2a Verbs (n=30):

- Pre-test norming of 168 verbs from Levin (1993)
- Evaluated on Valence/Conceptual (phys-psych) 7-pt likert scale
- 40 Verbs chosen based on distance to endpoint

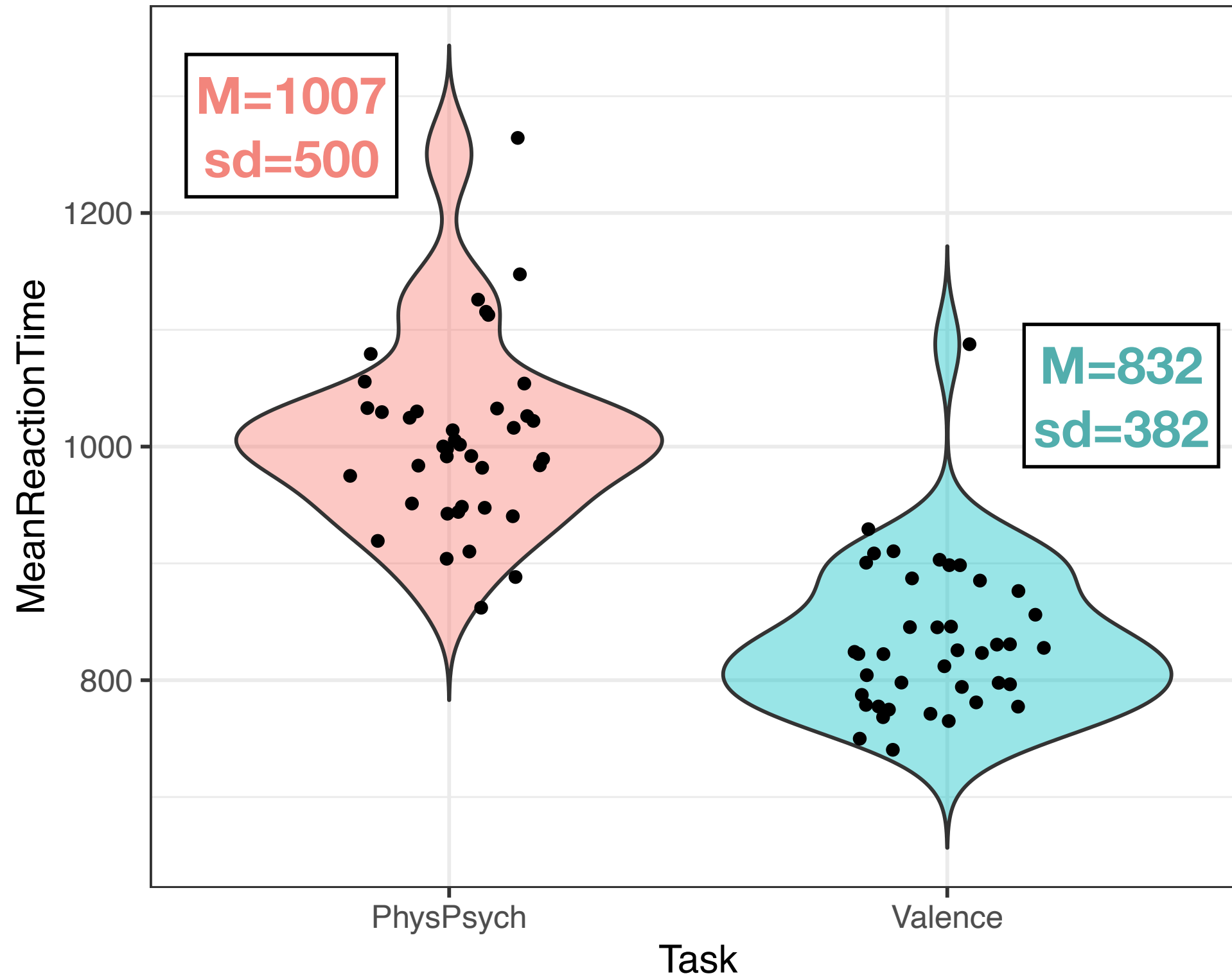
2b Adjectives (n=30):

- Pre-test norming of 289 adjectives from Warriner et al (2013)
- Evaluated Social/Physical on 7-pt Likert
- 40 Adjs chosen based on weighted average and minimized rating variance

Responses 2a



Exp 2a (Verbs) Results

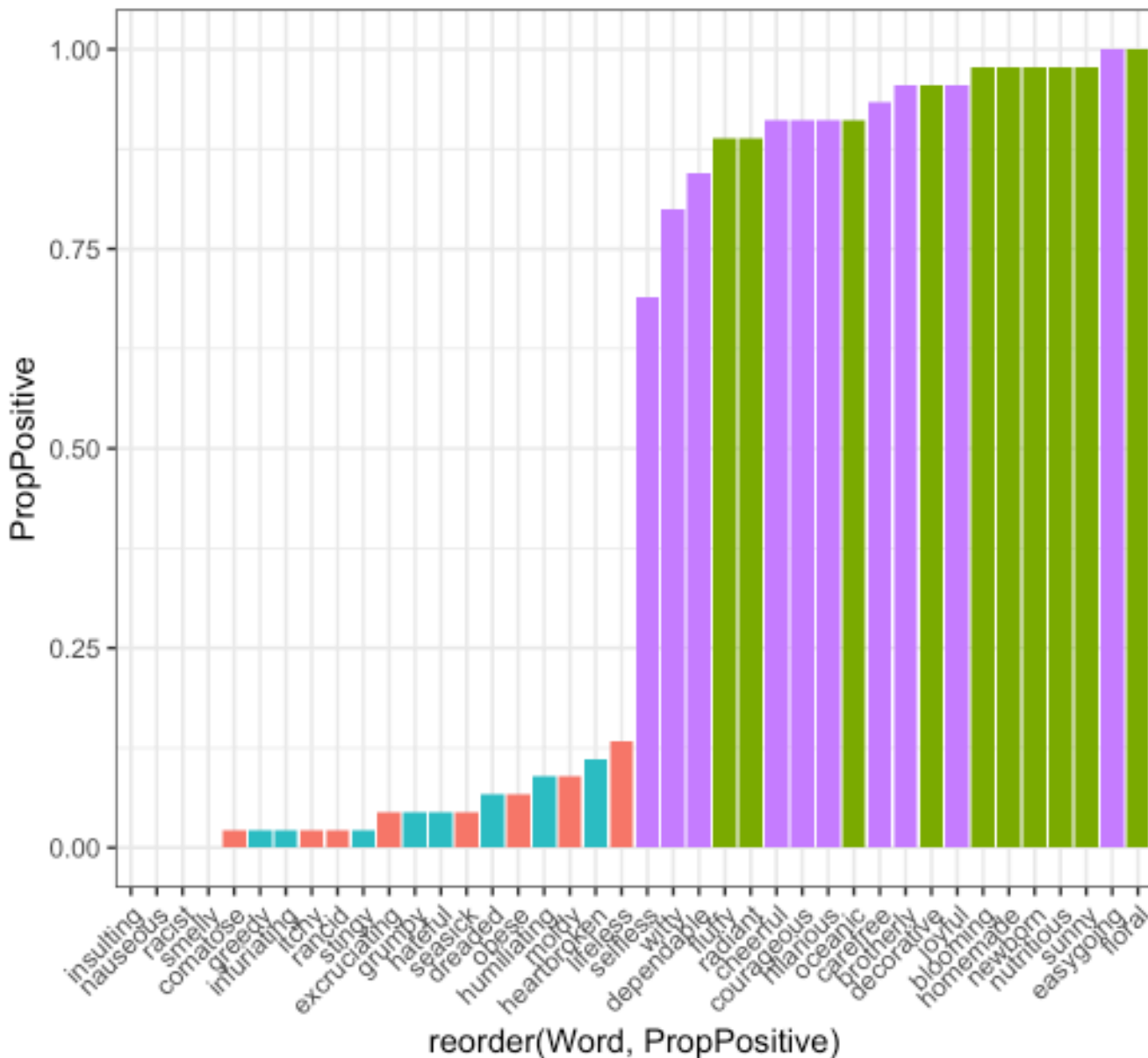


Verbs

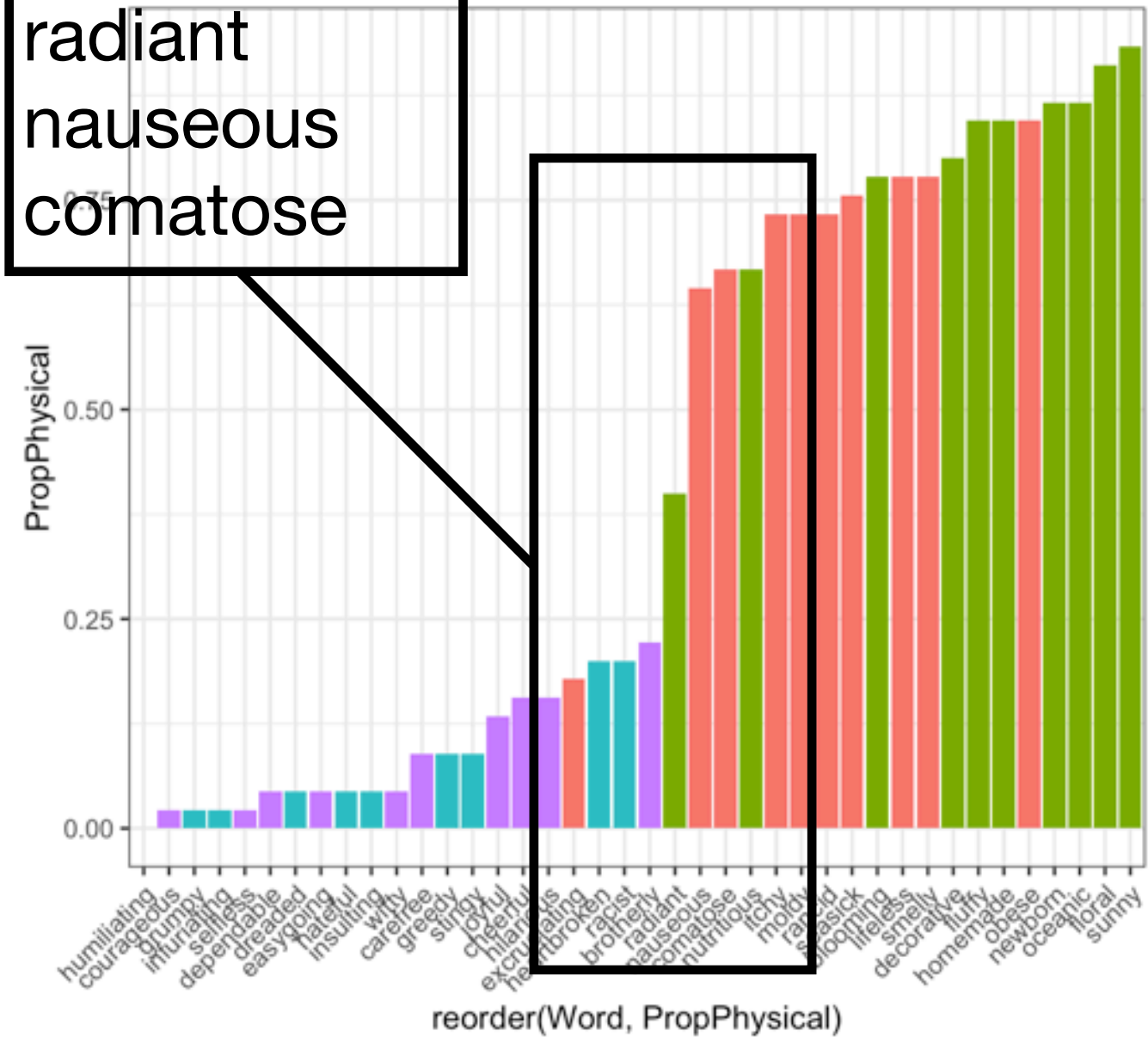


Responses 2b

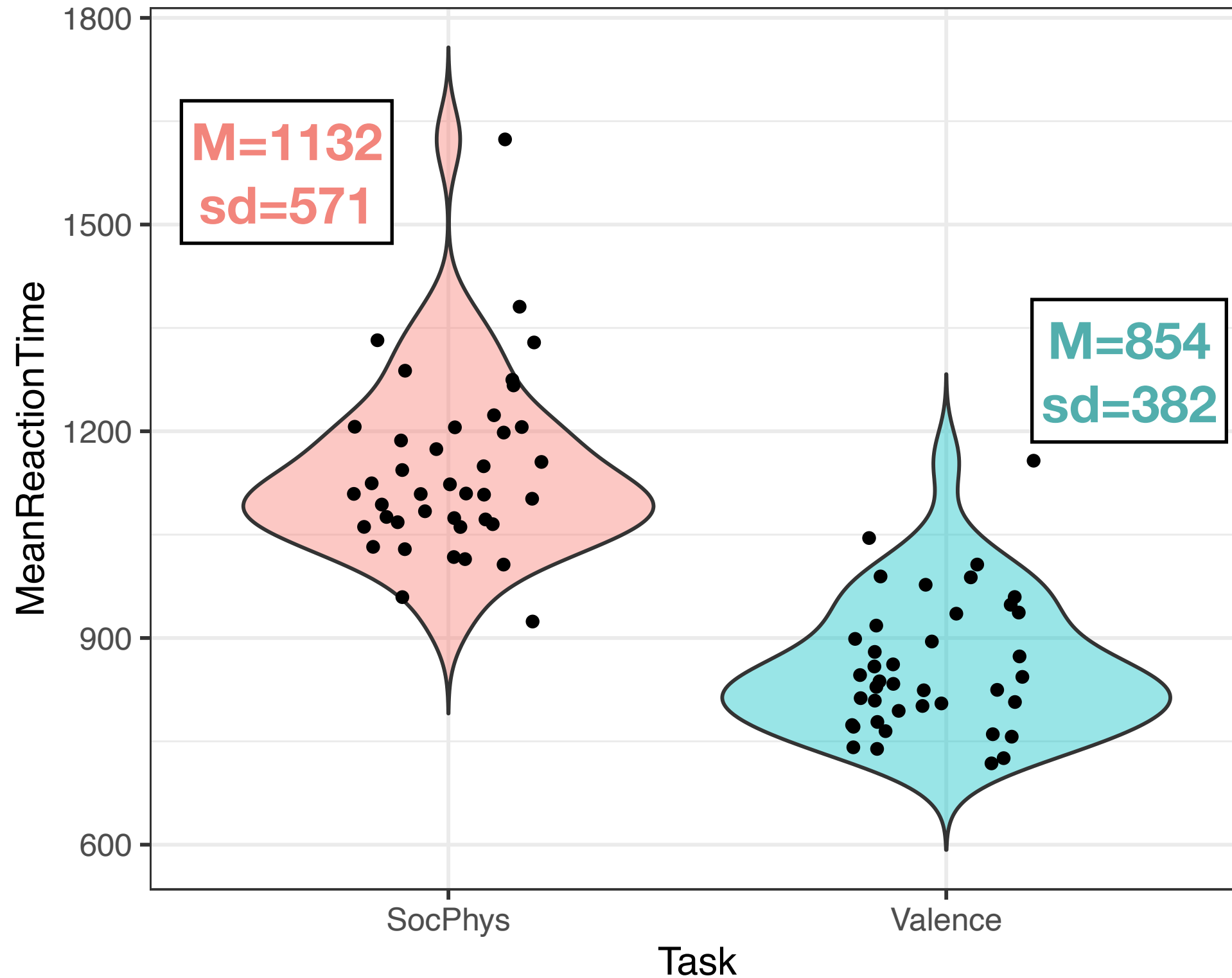
Feature Combo



excruciating
radiant
nauseous
comatose



Exp 2b (Adjs) Results



Adjs



Experiment 3:

Animacy (Nouns)

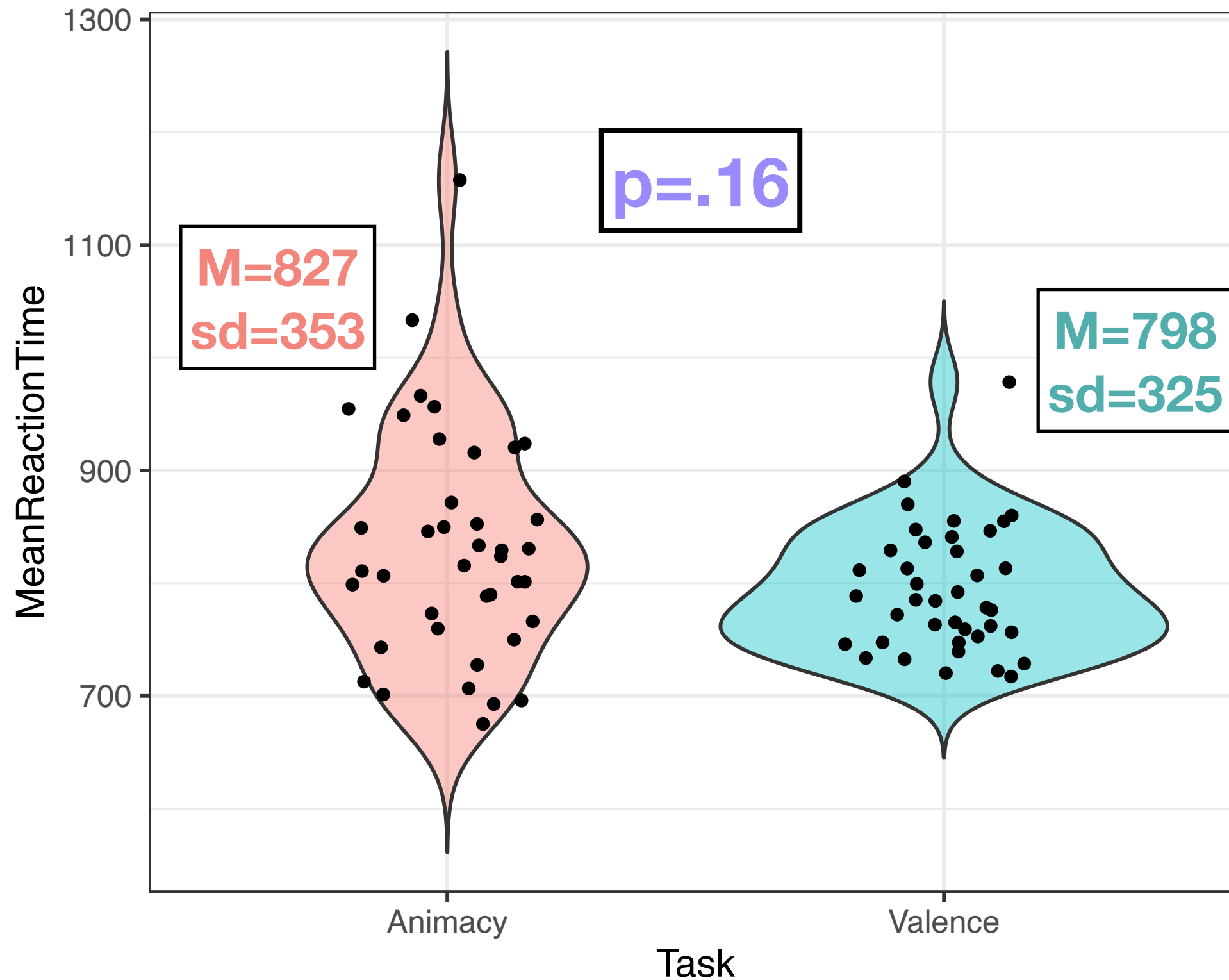
Exp 3 Stimuli

Using Corpus norms for Valence (Warriner et al 2013)

WordNet (Fellbaum 1998) words with 'living thing' or 'organism' as synset

High Frequency, and within ± 2 sd of mean Word Length

Exp 3 Results (Nouns)



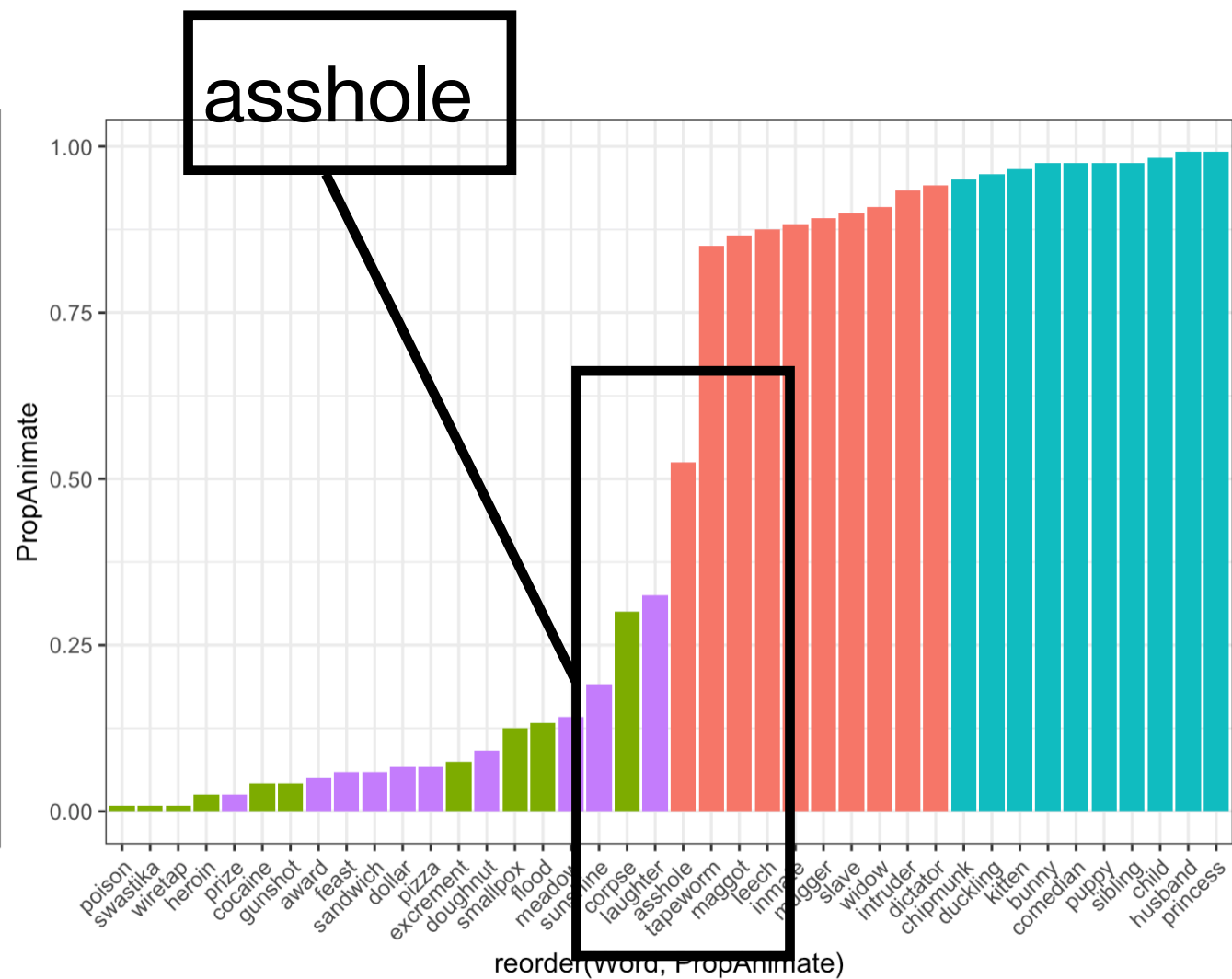
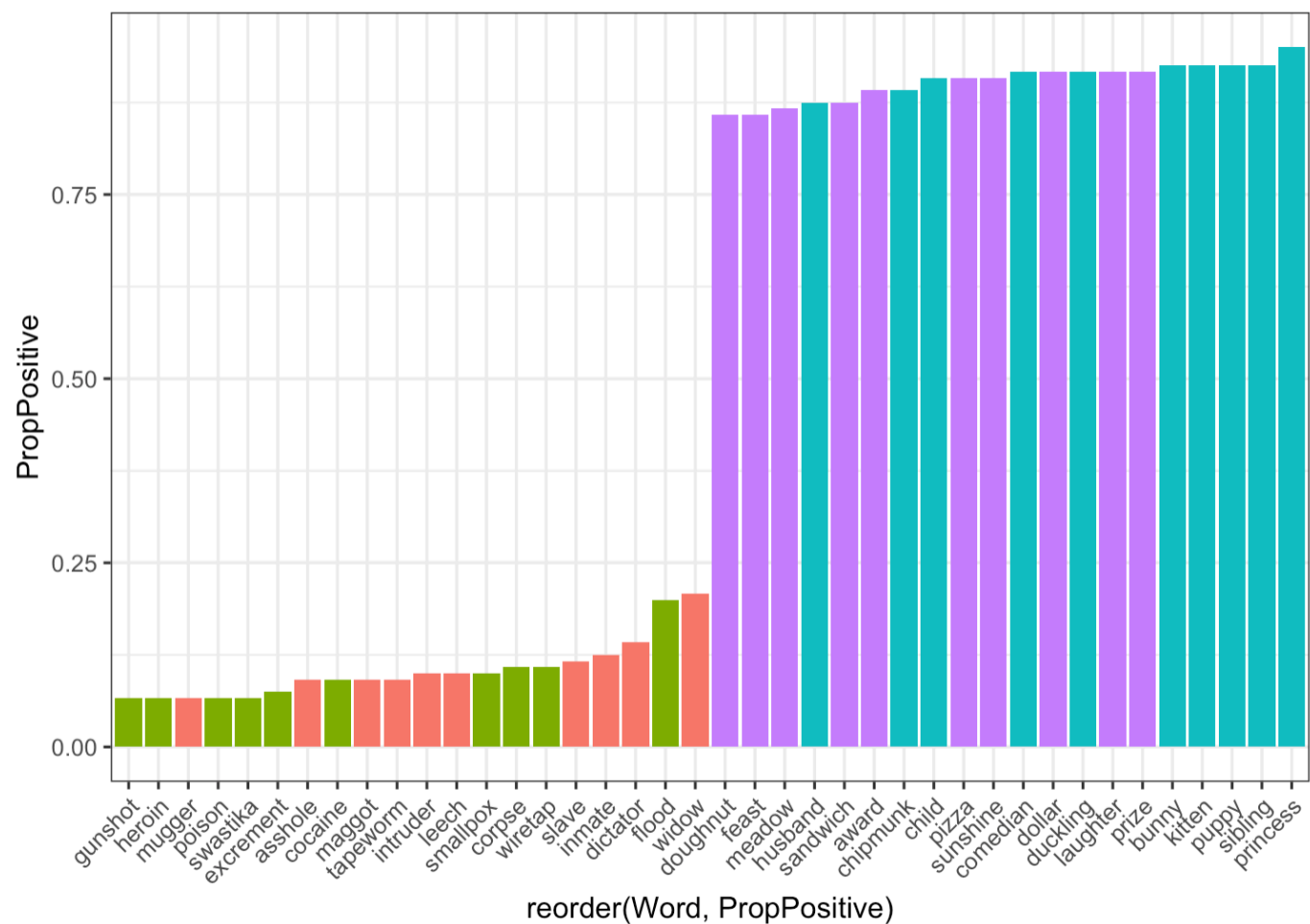
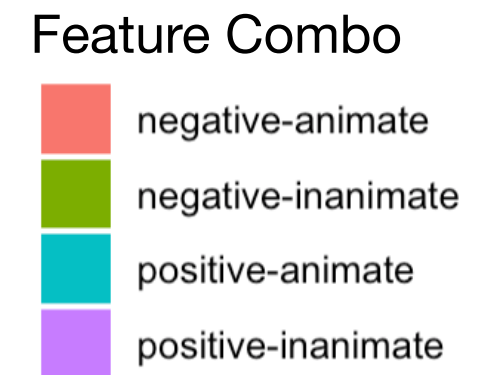
Nouns



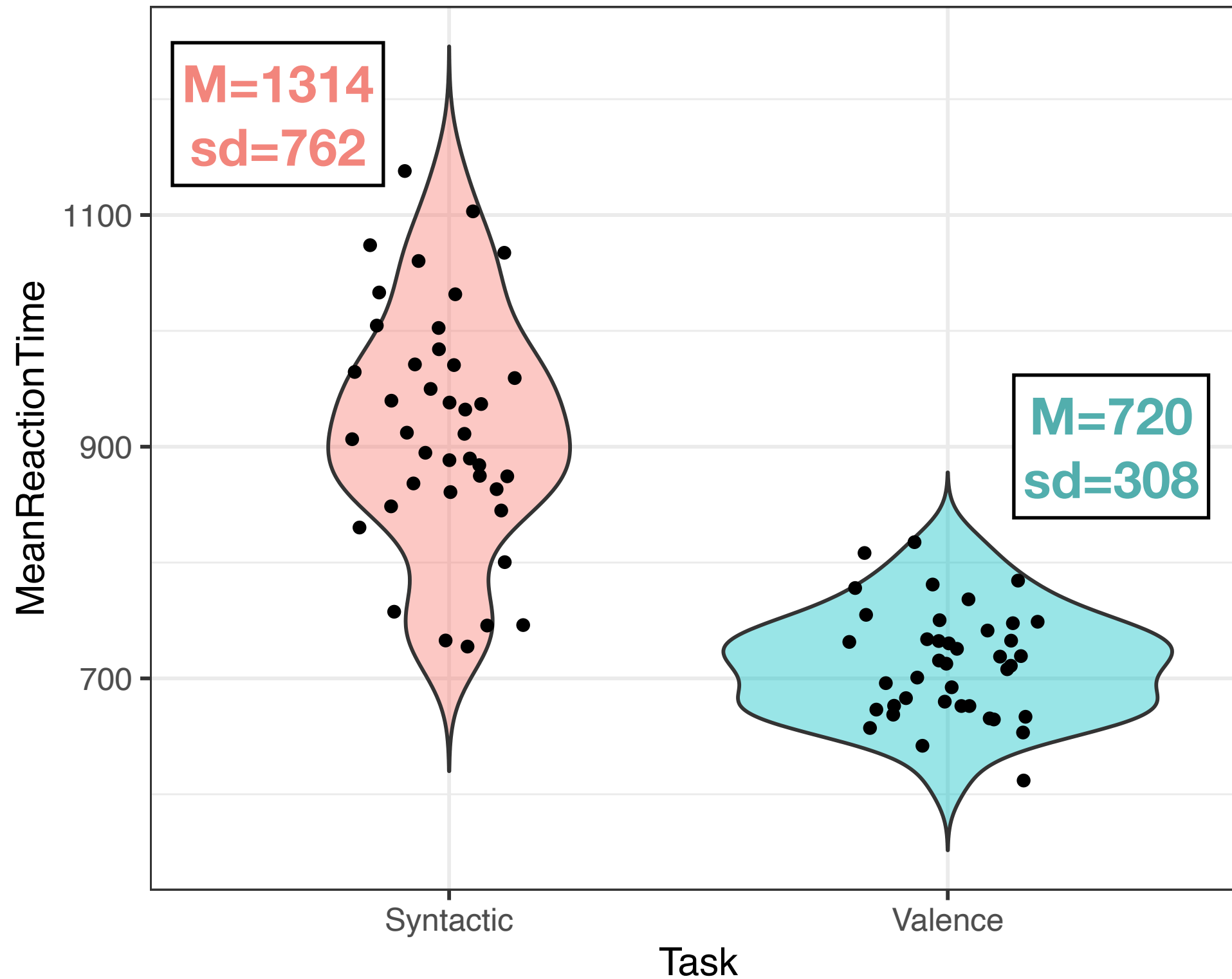
Experiment 4:

Syntax

Responses Exp3



Exp 4 Results



Verbs

Nouns

MeanReactionTime



Task



Responses Exp4

Feature Combo

